Unequal Division of Labor, Gender Role Attitudes, and Fertility Decisions *

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Abstract

This paper studies how the division of labor between spouses shapes fertility in Japan, a country with very low fertility and large gender gaps in paid and unpaid work. Using panel data on time use and gender role attitudes, I estimate a life-cycle model in which couples jointly decide on consumption, time use, and fertility. In the model, an unequal division of labor can be costly to women beyond forgone earnings, through perceived unfairness and lost career fulfillment. The estimates show that this cost falls mainly on progressive women. Cutting husbands' weekly hours from 60 to 54 raises completed fertility by up to 4.4%, most among progressive women, although husbands spend at most 14% of the freed time on housework. Under these hours cuts, a more equal division of housework raises both women's labor supply and fertility, suggesting that greater gender equality need not come at the expense of fertility.

Keywords: low fertility, domestic labor division, gender roles, work-life balance

JEL classification: D13, D79, J1.

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1 Introduction

Fertility has fallen below replacement level in nearly all high-income countries. A growing literature attributes much of this decline to a mismatch between women's expanding economic autonomy and the persistence of traditional gender roles (Goldin, 2025a,b). This mismatch is most visible within couples: women perform most housework and childcare while men specialize in market work, and the gap widens sharply after childbirth (Kleven et al., 2019, 2024; Aguilar-Gomez et al., 2026).[1] As women's education and career prospects rise, this division becomes increasingly costly for them, and many respond by having fewer children, or none.

These costs are not only pecuniary. Beyond forgone earnings, women may perceive an unequal division of labor as unfair (Lennon and Rosenfield, 1994; Hu and Yucel, 2018) and lose career fulfillment (Goldin, 2006). How heavily these costs weigh depends on women's own views about gender roles, and ignoring these costs understates the cost of children (Boneva et al., 2026). Yet the division also depends on men, whose time spent on home production is limited by gender norms and by the demands of their jobs. This constraint binds most tightly in East Asian countries, where fertility rates are among the lowest in the world and spouses' roles among the most unequal (Myong et al., 2021). Long and inflexible working hours keep men there out of childcare (Goldin, 2014; Nagase and Brinton, 2017). This paper asks two questions. First, how do women's gender role attitudes shape the costs they attach to an unequal division of labor, and for which women do these costs matter most? Second, can shorter working hours and a larger share of domestic labor for husbands raise fertility?

I study these questions in Japan, where fertility has been below replacement since the mid-1970s, and couples exhibit one of the largest gender gaps in paid and unpaid work among high-income countries (OECD, 2025). Successive family policies have barely raised fertility rates, partly because persistent gender norms and long, rigid working hours keep spouses' paid and unpaid work highly unequal (Boling, 2008; Nemoto, 2013). The Japanese Panel Survey of Consumers records both spouses' detailed time use and expenditure, together with women's gender role attitudes and birth intentions. Two empirical patterns emerge from the data. First, gender specialization intensifies after the first childbirth, and the differences across women holding different gender role attitudes widen. Second, among women who have no birth intention, progressive women increasingly attribute their decision to a lack of childcare help from their husbands as the housework gap grows. No such difference appears across education or wage levels.

[1] Part of this unequal division of labor reflects comparative advantage, since households gain from specialization when spouses differ in earning capacity (Becker, 1993). Yet comparative advantage alone cannot explain this division in advanced economies, where gender norms also play a primary role (Siminski and Yetsenga, 2022). For example, even wives who out-earn their husbands still do more housework (Bertrand et al., 2015).

The paper's main contribution is to develop and quantify a mechanism that models of efficient household specialization may overlook, the subjective cost of an unequal division of labor, which varies with women's gender role attitudes. Structural models of household time use and fertility typically treat the housework gap as efficient specialization driven by comparative advantage, with gender norms entering, if at all, as a penalty on women's market work. In such models, a more equal division of labor raises men's opportunity cost of children and need not raise fertility. I instead allow an unequal division of labor to carry a non-pecuniary cost or benefit, since women may see it as unfair or at odds with their gender views. This channel operates through two margins, the housework gap between spouses and women's long market hours, and the cost or value women attach to each varies across three attitude groups. This has two implications. First, in the model's simulations of shorter hours for husbands, a more equal division of labor is compatible with higher female labor supply while raising fertility. Second, responses differ across attitude groups in ways a binary measure of gender values would miss. For instance, shorter work hours for husbands raise fertility most among progressive women, but more among conservative than moderate women. A second contribution is to treat husbands' long hours as a constraint set by workplace norms rather than a career choice, which allows me to evaluate working-time policies such as Japan's 2018 Work Style Reform. Methodologically, the paper integrates home production, attitude-dependent costs, and endogenous fertility in a dynamic collective model estimated on both spouses' time use.

To quantify the channel and evaluate policies, I develop and estimate a life-cycle model in the spirit of Mazzocco (2007); Voena (2015); Foerster (2024). Married couples with distinct preferences cooperatively choose consumption, time spent in market work, home production, and leisure, and whether to conceive. Their bargaining power is set at marriage and stays fixed. Children raise the value of home production, which increases both spouses' housework time, especially women's. Women's time away from market work slows their human capital accumulation and lowers their wages. Husbands' hours are fixed at 60 per week, reflecting workplace norms, which allows me to simulate the effects of shortening them. Women also bear two non-pecuniary costs (or benefits) that depend on their gender role attitudes, one from doing more housework than their husbands, which may be more costly for progressive women, and one from working long market hours. The latter captures career fulfillment for progressive women and the stigma of departing from traditional roles for conservative women.

The main finding is that a more equal division of labor raises both female labor supply and fertility, because it reduces the non-pecuniary cost women attach to an unequal division of labor. The estimates show that this cost is borne mainly by progressive women, who value long market hours and dislike a disproportionate share of housework once they become mothers, while conservative and moderate women prefer a traditional division.

Three counterfactual results support this finding. First, gender role attitudes account for 60.2% of the fertility gap between moderate and progressive women and 33.7% of the gap between conservative and progressive women, with the largest fertility responses among women with the highest earning potential. Second, husbands' long working hours are a binding constraint on fertility. Cutting husbands' market hours by 10%, from 60 to 54 per week, raises completed fertility by 4.4% among progressive women, 3.7% among conservative women, and 1.3% among moderate women, even though husbands devote only 10–14% of the freed time to domestic work. This modest reallocation narrows the gap in childcare burden between spouses and allows women to increase their market hours by 6–20%, which lowers the non-pecuniary cost of an unequal division of labor and makes having more children more attractive. A reduction of three hours per week, matching the estimated effect of Japan's 2018 Work Style Reform, raises fertility by up to 2%. Third, when husbands devote half of the freed time to home production after their market hours are cut to 54 per week, completed fertility rises by up to a further 1%, while women's market hours increase by an additional 2–4%.

This last result speaks to a common concern that a transition toward more egalitarian gender roles may lower fertility, because households lose the gains from specialization and women's career costs of children rise. The results suggest that this need not be the case, as a more equal division of labor allows women to pursue a career and have more children at the same time.

Related Literature. This paper relates to three strands of literature. The first concerns low fertility in high-income countries. An important factor behind it is women's greater autonomy and a mismatch between men's and women's preferences over the traditional division of labor, including both home production and market work (Goldin, 2025a,b). While a heavy childcare burden on women lowers fertility intentions (Cardia and Gomme, 2018), the way spouses share this burden plays a distinct role.[2] Doepke and Kindermann (2019) find that an unequal division of childcare makes women more opposed than their husbands to having more children. When husbands contribute more to home production, women are more willing to have additional children and to participate in the labor market (Feyrer et al., 2008; De Laat and Sevilla-Sanz, 2011).

Unequal division of labor lowers fertility through two main mechanisms. The first is the child penalty, the effect of children on women's labor market outcomes (Eckstein et al., 2019; Kleven et al., 2019; Andresen and Nix, 2022; Kleven et al., 2024). As women's education rises, the time spent on childcare carries a higher opportunity cost, and the

[2]The division of labor between spouses also matters for household welfare beyond fertility. Spouses' housework time is substitutable, so households can absorb income shocks by reallocating it between partners, which serves as a consumption-smoothing channel (Blundell et al., 2018; Ma and Shi, 2020). Studies that focus solely on the size of the childcare burden therefore miss an important dimension of household welfare.

anticipated penalty shapes both fertility and career choices (Adda et al., 2017). The second is the non-pecuniary cost of unequal division of labor within the household. Women often perceive the paid and unpaid labor imbalance as unfair (Greenstein, 1996; Coltrane, 2000; Goldin, 2006), and such unfairness perceptions are associated with lower marital and family satisfaction (Lennon and Rosenfield, 1994; Hu and Yucel, 2018). Recent economic evidence points in the same direction. Sun et al. (2025) show that childbirth temporarily reduces women's paid work but persistently increases unpaid domestic work, raising their total workload and harming mental and physical well-being. These costs are well documented in sociology and psychology but rarely examined in economic studies of household decision-making. This paper incorporates this mechanism and uses time-use data to quantify its effect on fertility decisions.

The second strand concerns gender identity and economic outcomes. Following Akerlof and Kranton (2000, 2016), gender identity reflects prescriptions about how each gender should behave. When shared within a society, these prescriptions form gender norms and shape family decisions such as marriage, labor supply, home production, and fertility (Fernández and Fogli, 2006, 2009; Bertrand et al., 2015, 2021; Bursztyn et al., 2017), most visibly through a division of labor that assigns housework and childcare to women and most market work to men (Siminski and Yetsenga, 2022; Sakamoto and Kohara, 2025). In East Asia, such norms depress fertility because women bear a disproportionate share of childcare (Myong et al., 2021; Kim et al., 2024). Yet individuals within the same society hold different gender role attitudes. These attitudes shape matching, intra-household allocation, and maternal labor supply (Goussé et al., 2017; Mensinger and Pugnaghi Zimpelmann, 2025; Boneva et al., 2026), as well as how unfair women find an unequal division of labor (Greenstein, 1996; Coltrane, 2000) and how differently spouses care about the distribution of time within the household (Cosaert and Lu, 2026). Hwang (2016) shows that the conflict between women's attitudes and traditional norms helps explain falling marriage rates among educated women in East Asia. Existing work on fertility, however, treats norms as common to all women in a society. This paper shows that heterogeneity in attitudes within a society matters: the same division of labor imposes different costs on different women and leads to different fertility choices.

Methodologically, this paper builds on life-cycle models of household decision-making. Most of these models study marriage, divorce and intra-household allocation while treating fertility as exogenous (Voena, 2015; Fernández and Wong, 2017; Reynoso, 2024; Foerster, 2024), or study parental investment given exogenous fertility process (Del Boca et al., 2014). Fewer make fertility an endogenous choice. Adda et al. (2017) emphasize the trade-off between labor supply and fertility but abstract from home production, and Doepke and Kindermann (2019) model spouses' disagreement over fertility without modeling home production explicitly. Closest to this paper, Mattia (2026) estimates a dynamic collective model of time use, employment, and fertility for Italy, in which tra-

ditional gender values generate a utility penalty when women work full-time. My model departs from his in several respects. I do not restrict the sign of women's preferences over long working hours, so whether long hours generate utility or disutility depends on women's gender attitudes. I also model attitude-dependent preferences over the housework gap between spouses, again without restricting their sign. Together, these two terms form the non-pecuniary cost or benefit that varies across three attitude groups, so a more equal division of labor, in which women work longer hours and do less childcare, need not lower fertility. Finally, I fix husbands' hours to capture workplace norms, which are particularly rigid in Japan.[3] Building on Voena (2015), Lise and Yamada (2019), and Foerster (2024), this paper develops a dynamic household model that integrates home production with endogenous fertility decisions.

The remainder of the paper is organized as follows. The next section describes the institutional background in Japan. Sections 3 and 4 present the data and empirical evidence from event-study and interaction analyses. Section 5 develops the theoretical model, and Section 6 presents the estimation results and model fit. Section 7 conducts counterfactual analyses evaluating the effects of the gender-view channel and husbands' long working hours on fertility outcomes. Finally, Section 8 concludes.

2 Institutional Background

This paper uses Japan as an example to examine how the unequal division of labor within households contributes to low fertility, drawing on the country's cultural and institutional background. Japan has experienced total fertility rates below the replacement level since the 1970s. Although the government has introduced numerous pro-natal policies, their impact has been limited, partly because the imbalance between spouses remains unresolved (Jones, 2019). For instance, despite generous parental leave provisions, men's childcare leave uptake remained below 4% until 2016, compared with over 80% among women employed at childbirth (Ministry of Health, Labour and Welfare, 2025).

Japanese couples exhibit a large gender gap in the division of paid and unpaid labor. According to OECD (2025), Japanese women performed on average 82% of the total unpaid domestic work between spouses in 2023. Although labor force participation rates are relatively close (87% for men and 75% for women aged 15 to 64), about 34% of employed women work part-time. As a result, women's average paid working hours are roughly half those of men, with the disparity particularly pronounced among households with children (Lise and Yamada, 2019; Sakamoto and Kohara, 2025). Two features of

[3] Another strand of literature models endogenous fertility and child investment in a non-cooperative framework, such as Caucutt et al. (2002); Tartari (2015); Brown et al. (2025). This paper instead follows the cooperative approach of Chiappori (1988, 1992) and Chiappori et al. (2002), treating intra-household allocation as the outcome of coordinated decisions between spouses. This approach assumes Pareto-efficient allocations, a standard assumption in recent household models.

Japan contribute to this gap. Traditional gender norms remain widespread, with Japan ranking 118th of 148 countries in the 2025 Global Gender Gap Index, the lowest among G7 countries, and 112th on economic participation and opportunity (World Economic Forum, 2025). In addition, long working hours and rigid corporate hierarchies limit the time men can devote to childcare and other housework.[4]

At the same time, women's education and employment opportunities have expanded, gradually shifting women's gender role attitudes. Japan's 1990 education reform, for example, made women's gender views more egalitarian while leaving men's unchanged, and it also changed how spouses divide household tasks (Hara and Rodriguez-Planas, 2025). As a result, the tension between the traditional division of labor and women's more egalitarian expectations has become increasingly visible.

Long working hours arise from organizational features such as lifetime employment, seniority-based wages, and intense competition (Ono, 2018). Although the Labor Standards Act has set a limit of 8 hours per day and 40 hours per week since 1987, firms could easily bypass these limits through Article 36 Agreements. Once such agreements were signed by employees or their representatives, companies could legally require overtime with no binding cap or penalty. Under this regulation and a long-standing overwork culture, employees typically worked very long hours. In my sample, married men work about 60 hours per week including commuting time, about 10 hours per day on weekdays, and many also work on weekends (see the summary statistics in Section 3). Because men are typically the main breadwinners, these workplace norms, though applied to all employees, reinforce gender specialization within households (Nemoto, 2013).

To improve work–life balance and encourage women's labor force participation, the government enacted the Work Style Reform law in 2018.[5] From April 2019, the reform made overtime caps legally binding, limiting overtime to 45 hours per month and 360 hours per year in principle, with fines for employers that exceed them. Using survey data, Burdin et al. (2026) find that the reform cut total overtime by about three hours per week for men above the cap. I use this reduction in Section 7.2 to examine whether changes in husbands' work hours can reshape the household division of labor and fertility.

[4]See Brinton and Oh (2019) for a comparative study of postindustrial Japan and South Korea. Drawing on in-depth interviews with highly educated women and men, the study shows that market structures and workplace norms reinforce a highly gendered household division of labor, leading many married women either to withdraw from employment or to limit childbearing to one child. Related demographic studies include Boling (2008), Brinton and Lee (2016), and Nagase and Brinton (2017).

[5]The full name is "Act on the Arrangement of Related Acts to Promote Work Style Reform". The law was enacted on July 6, 2018.

3 Data

This study uses data from the Japanese Panel Survey of Consumers (JPSC), a longitudinal survey covering 1993–2021. The JPSC tracks female respondents aged 24 and above, including both single and married women, and married respondents also report information on their spouses.[6] The survey provides comprehensive demographic information, as well as respondents' gender role attitudes and stated birth intentions.

The JPSC is an annual panel of women, with new cohorts added periodically. The first survey, in 1993, covered 1,500 women aged 24 to 34. New cohorts of young women were added in 1997, 2003, 2008, and 2013, so that the panel covers a wide range of ages. Each cohort comprised about 125 respondents per year of age when first surveyed. Attrition between waves is low, while new cohorts refresh the sample. In years when new cohorts were added, between 2.6% and 5% of women left the panel after the previous wave, while new entrants made up 16% to 31% of the female sample. In 2013, for example, new entrants made up 16% of the sample, and only 2.7% of women left after 2012.

I use the waves from 1993 to 2019, before the COVID-19 pandemic. The sample includes married women and their husbands who meet four criteria: (1) their demographic characteristics, such as age, education, and work experience, are observed; (2) their expenditure and time-use variables are non-missing for at least one year; (3) at least one spouse earns a wage; and (4) the wife's gender role attitudes are observed in at least one wave. The resulting sample includes 1,758 households, with 18,899 household-year observations.

3.1 Intrahousehold allocation and demographics

One feature of the dataset is that it records detailed expenditure categories and time-use activities for each spouse and the entire household. Household expenditures are reported across five categories: spending on the entire household, the female respondent's expenses, the husband's expenses (if the female respondent is married), child-related expenses, and other expenses. I treat household spending as family public consumption, spouse-specific expenditures as private consumption for each spouse, and child-related items as expenditure on children.

Time use is recorded for commuting, work, schoolwork, housekeeping and childcare, leisure, and other activities including sleep, meals and bathing. Commuting and schoolwork are combined with work to form a generalized measure of market work hours. Housekeeping and childcare are combined to form home production hours, while leisure is retained as a measure of private leisure. Since the survey collects daily schedules separately for weekdays and weekends, I sum each activity across weekdays and weekends to

[6] Since same-sex marriage is not legally recognized in Japan, all marriages in the dataset are heterosexual.

construct weekly hours for each category.

Table 1 presents summary statistics for married women in the sample and their husbands, characterizing key features of Japanese households. The bottom panel reports basic demographic characteristics. On average, wives in the sample are 37.6 years old, while husbands are 39.8. Educational attainment is high in the sample, with the majority of both wives and husbands holding a tertiary degree. Among wives, 4.5% have a middle school education, 39.2% have a high school education, and 56.4% have attained tertiary education, including college, university, and graduate school. Among husbands, the corresponding shares are 8.9%, 38.7%, and 52.4%, so wives are slightly more likely than husbands to hold a tertiary degree.[7] At the household level, couples have 1.72 children on average, the average age of children is 9.7 years, and wives have their first child at 27.2 years on average.

The top panel of Table 1 reports average expenditures and expenditure shares for spouses' private consumption and child-related consumption. On average, 6.4% of household expenditure is reported as the wife's private consumption and 14.1% as the husband's, so husbands' private consumption is more than twice that of wives. Around 13.6% of expenditure falls under the child-related category, with the remaining 66% attributable to household public consumption.

Wives and husbands also differ substantially in their time allocation across market work, home production, and leisure. Wives spend around 23.9 hours per week on market work and 45.6 hours on home production, accounting for 14.2% and 27.2% of their total time, respectively. The corresponding figures for husbands are 62.1 and 8.0 hours per week, accounting for 37% and 5% of their time. Both spouses spend about 13% of their time on leisure. Other activities, such as sleep, meals, and bathing, are excluded from the analysis. These activities account for approximately 76 hours per week for wives and 75 hours for husbands, or roughly 45% of total time for both.

3.2 Gender role attitudes

The dataset records respondents' personal views regarding gender roles at home, which I use to construct a measure of individual gender role attitudes. The survey asks respondents a series of questions on family views, including their views on the division of labor between spouses.[8] Respondents rate their agreement with each statement. The two statements are:

"*Men should work outside the home and women should work at home*"

[7] Compulsory education in Japan consists of nine years, comprising six years of elementary school and three years of junior high school. Students are not required to pass a graduation or qualification exam to receive a junior high school diploma.

[8] The survey includes two other questions on marriage and divorce. Since this paper focuses on gender role attitudes regarding labor division within marriage, I use only the two questions related to market and domestic labor provision in the household.

Table 1: Summary statistics of married couples

	Mean [standard deviation]		
	Wife	Husband	Household
Expenditure, 1000 yen per month			
Private consumption	16.06 [22.73]	33.93 [29.56]	
(% of household total)	(6.40%)	(14.05%)	
Expenses on children			38.43 [68.27]
(% of household total)			(13.63%)
Time use, hours per week			
Market work	23.88 [22.29]	62.11 [15.05]	
(% of own time)	(14.22%)	(36.97%)	
Home production	45.62 [26.88]	7.98 [9.48]	
(% of own time)	(27.16%)	(4.75%)	
Leisure	21.97 [17.00]	22.77 [14.30]	
(% of own time)	(13.08%)	(13.56%)	
Demographics			
Age	37.55 [7.20]	39.80 [8.02]	
Middle school (%)	4.47 [20.67]	8.89 [28.47]	
High school (%)	39.16 [48.81]	38.71 [48.71]	
Tertiary education (%)	56.37 [49.59]	52.39 [49.94]	
Age of first birth	27.17 [4.27]		
Number of children			1.72 [1.01]
Average age of children			9.65 [6.96]
Age of the youngest child			8.02 [6.75]

Notes: The sample comprises 1,758 households and 18,899 household-year observations. Market work includes commuting and study time.

"*Mothers should not have jobs and should focus on childcare until their children are about three years old*"

Responses are recorded on a four-point scale: 1 = *I agree*, 2 = *I somewhat agree*, 3 = *I somewhat disagree*, and 4 = *I disagree*. I sum responses to the two questions to construct a Gender Value Index (GVI), with higher values indicating more progressive gender views. The index ranges from 2 to 8, yielding seven distinct values.

The survey has included questions on gender roles since 2011, covering the eleven waves up to 2021. Gender role attitudes are largely stable within individuals. Figure A1 shows a gradual aggregate shift toward more egalitarian views, with the average GVI rising by about 0.57 points between 2011 and 2021. This shift combines individual change with changes in sample composition, since a new cohort was added in 2013, when new entrants made up 16% of the female sample and only 2.7% of women left after 2012. Within individuals, the index rises by only about 0.02 points per year (Table A1), about 35% of the aggregate shift, so the rest reflects changes in sample composition. This drift is small relative to differences across women, as about two thirds of the variance in reported GVI is between women (Table A2). Attitudes also do not respond to the birth of a child (Table A2), and in an event study around the first birth, the estimated average of each attitude group stays within the range that defines it (Figure A2).

I therefore treat attitudes as a fixed individual trait and measure them by each woman's average GVI across all available waves. This smooths out wave-to-wave reporting noise and allows me to use information from all panel waves, including those in which gender roles were not asked. Respondents who never answered these questions are excluded. Using each woman's first or last reported GVI instead leaves about 92% of women in the same group, and nearly all reclassifications are to an adjacent group. Figure A3 shows the distribution of this average GVI for married women in the sample, which has a mean of 5.4. Based on this measure, I divide women into three groups, which I use throughout the subsequent empirical analysis and the structural model. Women with an average GVI below 4 are classified as conservative, those from 4 to below 6 as moderate, and those at 6 or above as progressive. Under this classification, 16.0% of women are conservative, 46.8% are moderate, and 37.2% are progressive.

Respondents' gender role attitudes are largely independent of education and wages. Individuals with higher education and greater earning capacity tend to be more progressive, but the association is weak. A regression of GVI on educational attainment and wages produces a very low R-squared of 0.041, as shown in Table A3. A similarly weak association between gender views and education or wage is reported by Goussé et al. (2017) for Danish households. In addition, Figure A4 presents the share of women with different gender role attitudes at each education level. Women with higher education tend to be more progressive on average, but the differences across education levels are small, and the relationship is not linear. For instance, among women with a tertiary

education, 16.7% are classified as conservative—a higher share than among those with a high school degree. This suggests that gender role attitudes are not strictly determined by education, calling for further study of individual gender views and behaviors beyond observable characteristics.

3.3 Stated fertility intentions

The dataset also records women's stated birth intentions and the reasons behind their childbearing decisions. In Section 4.3, I use the reasons given by women with no birth intention to examine whether women respond to the division of labor beyond its economic costs. Respondents are first asked the following question:

> "*Do you want to have a child or another child?*"

with three response options: 1 = *Yes, at any cost*; 2 = *Yes, with condition(s)*; and 3 = *No*. In the pooled sample, 24% of respondents report "*Yes, at any cost*", 18% report "*Yes, with condition(s)*", and 58% report "*No*".

Stated fertility intentions align closely with subsequent fertility decisions.[9] Among respondents who report "*Yes, at any cost*", 44% have another child within three years, compared with 18% among those who report "*Yes, with condition(s)*" and only 4.7% among those who report "*No*". This close correspondence between stated intentions and fertility outcomes indicates that self-reported intentions serve as reliable measures of childbearing willingness.

Depending on their stated intentions, respondents are asked one of two follow-up questions. Those who report a conditional birth intention are asked about the "*conditions for giving birth*", while those with no intention to have another child are asked about "*reasons for not having children*". These conditions and reasons, along with the percentage of respondents listing each factor, are presented in Tables A4 and A5.

Among the conditions for having another child, "cooperation from husband and family" is the second most frequently cited condition, cited by 45% of respondents. By contrast, continuing to work and access to external childcare services appear less consequential, each cited by roughly 23% of women. This suggests that the heavy childcare burden—particularly the husband's lack of involvement in childcare—is a major concern shaping women's fertility intentions. This concern is not offset by external childcare support, such as daycare centers. These findings point to the need to examine the division of childcare time between spouses, and its effect on fertility willingness.

[9] Tan and Tey (1994); Thomson (1997) document a similar correspondence between women's stated fertility intentions and subsequent fertility outcomes.

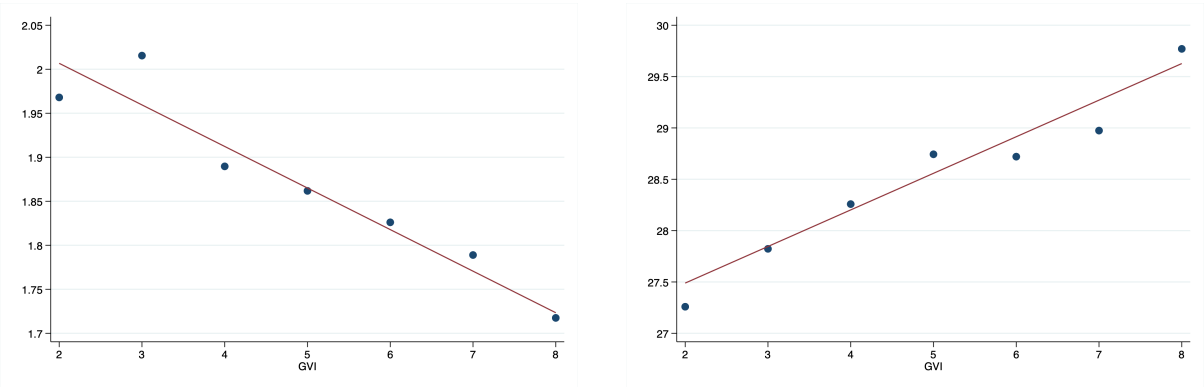
4 Empirical Evidence

This section presents empirical evidence that motivates the model's setup, organized around three findings. First, women with more progressive gender role attitudes have fewer children and delay their first birth. Second, a gendered division of labor is prevalent in Japanese households and intensifies after childbirth. Although progressive women share housework more equally with their husbands, they still face a substantial career penalty after childbirth. Third, among women who have no birth intention, progressive women more often attribute their decision to a lack of childcare help from their husband as the housework gap grows, while no such difference appears across education or wage levels.

4.1 Fertility outcomes and gendered division of labor

Women with higher GVI tend to have fewer children over the life cycle and to delay their first birth. Figure 1 plots fertility patterns against GVI, with higher values indicating more progressive gender role attitudes.[10] Panel (a) shows the completed fertility, measured as the maximum number of children a woman reports over the sample period. Panel (b) shows the average age at first birth. The sample is restricted to women aged 35 and above, 95% of whom have had at least one child. Completed fertility falls from about 2.0 children at a GVI of 2–3 to 1.8 at a GVI of 8, and the age at first birth rises from about 27 to almost 30.

This negative relationship between completed fertility and reported GVI persists after controlling for women's age and education (Figure A6). Table A8 reports the corresponding slopes of the fitted line, estimated on individual data, with and without these controls.



(a) Average number of children

(b) Women's average age at first birth

Figure 1: Fertility outcomes by Gender Value Index (GVI)

Progressive women devote more time to market work and less time to home produc-

[10] GVI here is each woman's own reported value, the sum of her answers to the two gender attitude questions, and takes integer values from 2 to 8. Each dot is the mean outcome for women with a given GVI value; the number of women per value is reported in Table A7.

tion, leading to a relatively more balanced division of domestic responsibilities. Husbands' market work, by contrast, remains largely stable at about 60 hours per week, regardless of their wives' gender role attitudes or the presence of children (Figure A7). The division of labor remains highly unequal even when wives earn more. Wives do more than 84% of home production even when they out-earn their husbands (Figure 2), about 20 percentage points more than comparable estimates for the United States and Australia (Stratton, 2020). This persistently high share cannot be explained by comparative advantage between spouses alone.[11] Instead, it is consistent with social norms shaping spouses' roles within the household (Siminski and Yetsenga, 2022).

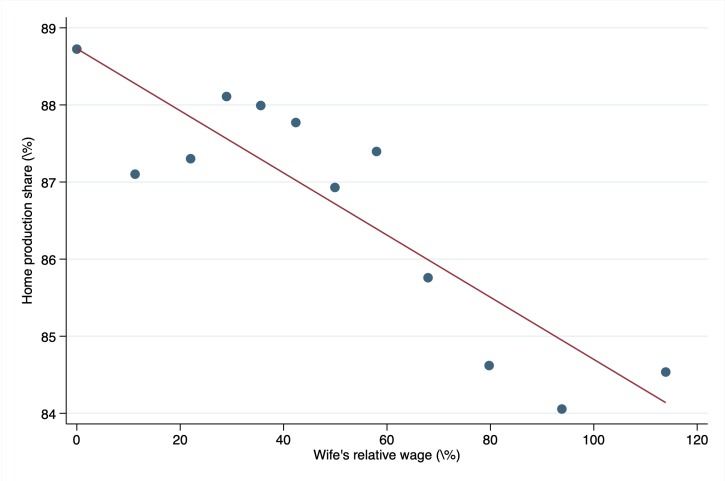


Figure 2: Wife's share of home production by relative wage

Notes: The wife's share of home production is her weekly home production hours divided by the total of both spouses. Relative wage is the wife's hourly wage divided by her husband's. Non-working wives are assigned a wage of zero, so a relative wage of zero identifies wives out of the labor market. Households in which the husband has zero earnings, about 1% of the sample, are excluded.

4.2 Child penalty

The highly unequal division of labor between spouses results in a heavier child penalty for women, which in turn shapes their fertility considerations. I assess the impact of the first childbirth on the evolution of spouses' time allocation using an event-study design following Kleven et al. (2019, 2024).

Women are assigned to one of three groups—conservative, moderate, and progres-

[11] Comparative advantage refers to differences in spouses' wages and earning capacity. The household allocation in this paper is efficient given spouses' preferences, including those tied to women's gender role attitudes. Myong et al. (2021) document similar departures from comparative advantage in South Korean families using a cooperative household framework; given the similar gender norms and time-use patterns, comparable departures are likely in Japanese households.

sive—and I estimate the following event-study regression separately for each group:

$$y_{it}^{g} = \sum_{k=-2,\, k\neq 0}^{6} \alpha_{k}^{g} \cdot D_{it}^{k} + \beta^{g} \cdot X_{it} + v_{it}^{g}, \tag{1}$$

where y_{it}^{g} is the outcome of individual i in group g in year t, and D_{it}^{k} equals one if year t is k years from the individual's first birth. The event window spans two years before and six years after the first birth, with the birth year ($k = 0$) omitted, so the coefficients α_{k}^{g} measure the change in the outcome at event time k relative to the year of the first birth. X_{it} controls for age, education, industry, and region and year fixed effects.[12] Standard errors are clustered at the individual level.[13]

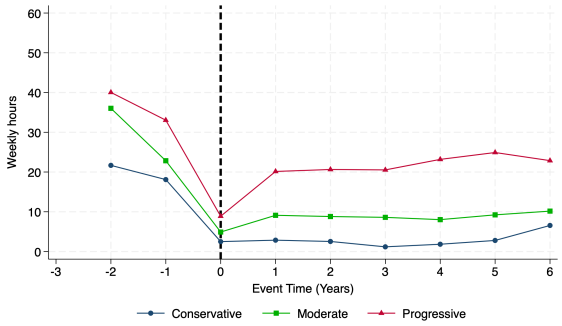
Figure 3 plots the group-specific estimates across event time.[14] The design allows for subsequent births in later years, and more conservative women are more likely to have additional children. To address the concern that differences across groups are driven by the number of children rather than gender role attitudes, I additionally report the same effects for the subsample of households with only one child during the sample period in Figure A8.

Comparing wives' outcomes across gender-attitude groups in Panels (a), (c), and (e), all groups of women experience reduced labor market involvement and lower wages after childbirth, and the gap between groups widens substantially—suggesting that childbirth reinforces women's roles in line with their personal attitudes. Panel (a) shows that progressive and moderate women are more likely to return to work after the first birth, though their labor supply fails to recover to pre-birth levels. Both groups maintain near full-time hours in the two years before childbirth, then shift to part-time work afterward, with progressive women averaging 20 hours per week and moderate women averaging 10 hours per week. Conservative women are the most likely to exit the labor market entirely, with average work hours falling below 5 hours per week. Panel (c) shows that women's home production hours increase sharply after the first birth and decline gradually thereafter; the gap between groups also widens relative to the pre-birth period, with home production stabilizing at 52 hours for progressive women, 60 hours for moderate women, and 66 hours for conservative women by the time the child turns six. A more direct measure of the career penalty is the hourly wage rate. Here I report the average wage among women who have income, so the trend reflects the wage penalty

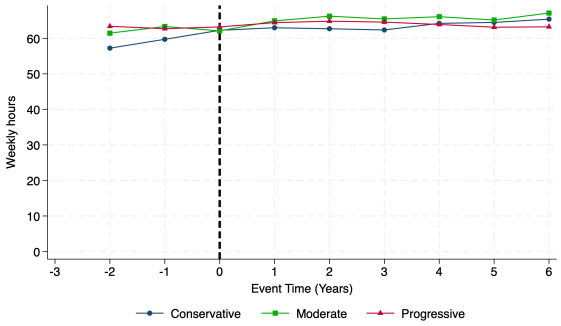
[12] For time-use outcomes, I control for both spouses' age and education; for wage outcomes, I control only for the individual's own age and education. Industry refers to the individual's own working sector, while region and year fixed effects are common to both members of the same household. Following Kleven et al. (2019, 2024), I do not include individual fixed effects; since all women in the sample eventually have a first child, identification comes from variation in the timing of first births conditional on the controls, and the flat pre-birth coefficients support this assumption.

[13] Table A9 reports the number of women by group and event time. The panel is unbalanced.

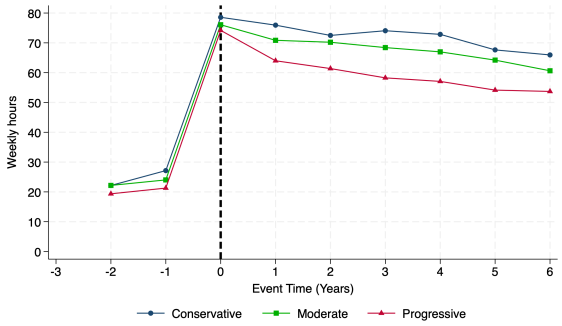
[14] The figures plot $\hat{\alpha}_{k}^{g} + \hat{E}[y_{it}^{g} \mid k = 0]$, the estimated coefficients plus the mean of the outcome in the birth year, to facilitate interpretation.



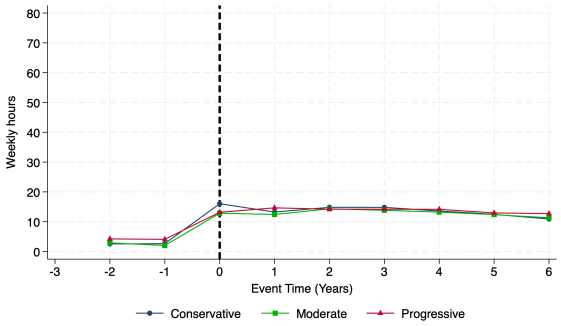
(a) Females' market work hours



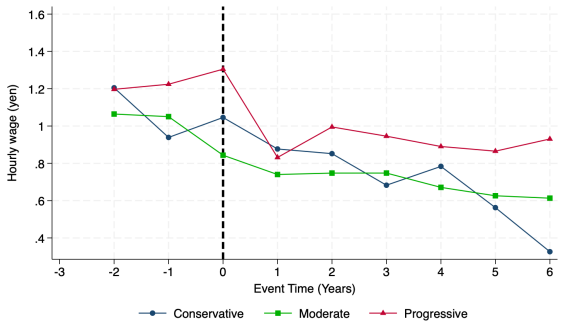
(b) Males' market work hours



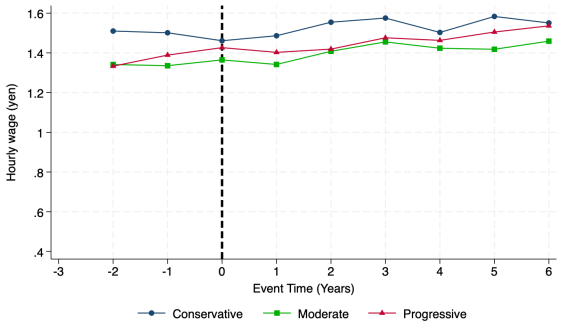
(c) Females' home production hours



(d) Males' home production hours



(e) Females' hourly wage rate



(f) Males' hourly wage rate

Figure 3: Effect of first childbirth on spouses' time allocation

among those who remain employed. All groups experience declining wages following childbirth, with the decline steepest among conservative women, indicating a more severe penalty for this group. Progressive women earn more on average than the other groups, though conservative women sometimes earn more than moderate women—suggesting that conservative women's earning potential is comparable to that of moderate women, and that their reduced labor supply reflects their personal values rather than differences in earning capacity.

Panels (b), (d), and (f) show outcomes for husbands. Husbands whose wives belong to different gender-attitude groups show little variation in their time-use patterns. Their market work hours remain stable at approximately 60 hours per week across the birth of the first child. Although husbands increase their home production hours for childcare, the spousal gap in home production widens substantially nonetheless. Panel (f) reports the hourly wage rate among husbands who are employed. Unlike women, men do not experience a significant wage decline after childbirth, showing no evidence of a career penalty. Husbands of conservative women earn more on average than husbands in the other two groups.

As a robustness check, I plot child penalties across spouses with different levels of educational attainment in Figure A9, to assess whether education—rather than gender role attitudes—better explains these patterns. The effects of childbirth by education group are far less pronounced than those by gender role attitudes, reinforcing that personal values regarding gender roles are a stronger predictor of women's career-versus-housework preferences than education.

4.3 Housework gap and stated birth intention

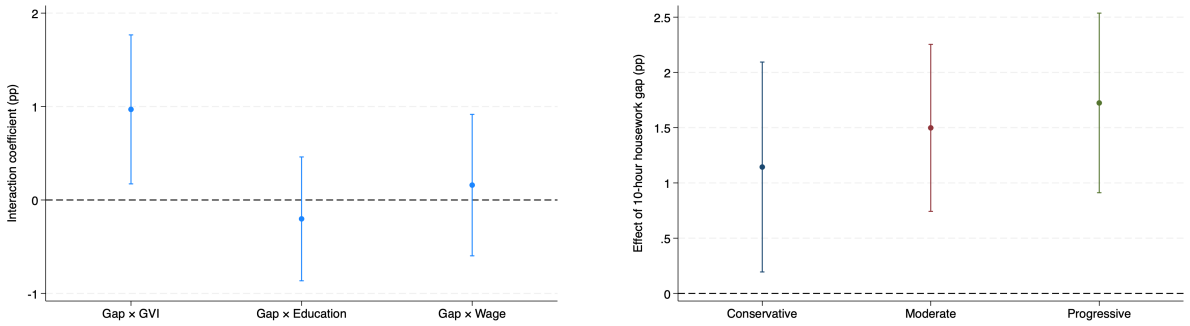
This section uses the stated reasons for having no birth intention to examine whether women respond to the division of labor beyond its economic costs. Unlike realized fertility, stated intentions reduce concerns about reverse causality, since children themselves increase housework.

Among women with no intention to have another child, the variable of interest is whether a woman cites the reason "I cannot expect that my husband will help me with childcare", which identifies women who do not intend to have another child because of their husband's limited involvement in domestic work. I examine whether a larger housework gap between spouses is associated with this reason, and whether the association differs with the wife's gender value index, education, and wage. Specifically, I estimate the linear probability model

$$y_{it} = \beta\, Gap_{it} + \sum_j \gamma_j\, Gap_{it} \cdot S_{ijt} + \sum_j \delta_j\, S_{ijt} + \boldsymbol{\beta}_c' \boldsymbol{X}_{it} + u_{it}, \tag{2}$$

where Gap_{it} is the wife's minus the husband's weekly housework hours, in units of 10 hours.[15] The characteristics of interest, S_{ijt} for $j \in \{\text{GVI}, \text{education}, \text{wage}\}$, are the wife's gender value index,[16] education, and hourly wage, each standardized to have mean zero and unit standard deviation.[17] y_{it} equals 100 if woman i cites the reason in survey year t and 0 otherwise, so the estimates are expressed in percentage points. The control variables $\boldsymbol{X}_{it}$ include year, wife's age, husband's education, and region fixed effects, as well as the number of children living at home. Each interaction coefficient γ_j measures how the association between the housework gap and the outcome changes with a one standard deviation increase in characteristic j, holding the other factors fixed. Comparing the three coefficients shows which characteristic is most strongly associated with women's aversion to an unequal division of housework among those with no birth intention. Since women are observed repeatedly across survey years, standard errors are clustered at the individual level.

Figure 4, panel (a), plots the three interaction coefficients, and Table A6 reports the full estimates, including those for the other conditions and reasons discussed below. Gender role attitudes are the only characteristic with a significant and positive interaction effect, while the interactions with education and the wage are insignificant. As the housework gap grows, progressive women are increasingly likely to cite an unsupportive husband as the reason for not intending another child, while women's response does not vary with their education or wage. This provides suggestive evidence that progressive women are more averse to an unequal division of labor when considering another child.



(a) Interaction coefficients **(b)** Effect across gender role attitudes

Figure 4: Housework gap and reasons for no birth intention

Notes: The sample is women with no birth intention. The outcome is citing an unsupportive husband as a reason, in percentage points, and the housework gap is measured in 10 hours per week. Bars show 95% confidence intervals, with standard errors clustered by individual.

To show the size of these effects, panel (b) plots the effect of the housework gap separately for the three attitude groups, estimated by interacting the gap with group

[15] Negative values, where the husband does more housework, account for 1.2% of observations.

[16] The GVI used here is the woman's reported GVI averaged across survey years.

[17] Standardizing places the three characteristics on a common scale, so their interaction coefficients can be compared directly.

indicators. The effects are significant in all three groups and largest among progressive women. A 10-hour larger weekly gap raises the probability of citing an unsupportive husband by 1.8 percentage points among progressive women, compared with 1.5 and 1.1 percentage points among moderate and conservative women.

As a comparison, Figure A5 shows the interaction coefficients for other outcomes. These include the reasons "childcare burden and education costs" and "continue working" among women with no birth intention, and the conditions "supportive husband", "continue working", and "childcare centers" among women with a conditional birth intention. Gender role attitudes and wages do not significantly change the response to the housework gap for any of these outcomes. Instead, more educated women respond more strongly to the gap through concerns about childcare and education costs and about continuing to work. This contrast suggests that attitudes shape women's response through spousal support specifically, whereas education shapes it through economic and career considerations.

5 Model

This section describes a dynamic collective model that characterizes spouses' intra-household allocation and fertility decisions. The model has three main features. First, couples make decisions collectively under full commitment, with the Pareto weight determined by their initial wage distribution. Second, fertility is an endogenous discrete choice made in each period. Third, the female spouse incurs a penalty for unequal home production hours between spouses.

The model is set in discrete time, where each period represents two calendar years. Females are 24 years old at the initial period, married, and endowed with an initial distribution of number of children associated with their gender-role views. In each period, spouses maximize the expected discounted sum of utilities over the remaining part of their life by choosing private consumption and household saving, and by allocating time between market work, home production, and leisure. In periods when the female spouse is fertile, the couple also decides whether to try to conceive an additional child.

The two spouses $i = 1, 2$ decide on the intra-household allocation of consumption and time use. I denote spouses' private consumption by c_t^i and joint saving assets by a_t. Each spouse's time endowment is divided among market work h_t^i, home production k_t^i, and private leisure l_t^i. For the fertility decision, I let a binary variable $d_t = 1$ denote that the spouses choose to conceive a child at period t, and $d_t = 0$ otherwise. Whether a child successfully arrives in the subsequent period is probabilistic and depends on the female spouse's age.

Spouses make decisions taking a range of state variables into account, collected in $\Omega_t = (\mu_t, a_t, K_t^1, K_t^2, n_t, o_t, age_t^1, z_1, \epsilon_t^1, \epsilon_t^2, \eta_t^C, \eta_t^{NC})$, where a_t is saving assets carried over

from the previous period, K_t^1 and K_t^2 are the human capital of female and male, n_t is the number of children living at home, o_t is the age of the youngest child, age_t^1 is the age of the female spouse, and z_1 is the female spouse's gender-role attitude, which is constant over time. In addition, ϵ_t^1 and ϵ_t^2 denote wage shocks to the spouses, and η_t^C and η_t^{NC} are the spouses' preference shocks toward conception and non-conception, respectively.

Married couples jointly make household decisions over the horizon. I follow the spirit of the collective model, which assumes spouses have distinct preferences and that the household maximizes the weighted sum of both spouses' utilities, with the weights representing their relative bargaining power. The resulting allocation is Pareto efficient (Chiappori, 1988, 1992), and more closely aligned with the preferences of the spouse who holds greater bargaining power. I assume the spouses are married prior to the initial period. Spousal bargaining power is determined at the beginning of the marriage, depends on the spouses' relative wage rates and remains constant thereafter. The model thus follows a full-commitment setting in which spouses agree on the Pareto-efficient allocation at the initial period of marriage, and outside options such as divorce or other shocks do not lead to renegotiation of bargaining power (Mazzocco, 2007; Theloudis et al., 2025).

I impose two core assumptions in the model. First, to reflect Japan's overwork culture and the limited variation in men's long working hours across work and childbirth related characteristics, male spouses' market work hours are fixed at 60 hours per week. Second, divorce is an absorbing state with no remarriage allowed [18], and the value of divorce is set very low, making the outside option of marriage unattractive relative to remaining married. [19]

Preferences. A household's flow utility in period t is a weighted sum of the spouses' individual utilities,

$$\begin{aligned} &U(c_t^1, c_t^2, l_t^1, l_t^2, k_t^1, k_t^2, h_t^1, h_t^2, d_t; n_t, o_t, K_t^1, K_t^2, z_1) = \\ &\mu_t^1 \, u^1(c_t^1, l_t^1, h_t^1, k_t^1, k_t^2, d_t; n_t, o_t, K_t^1, z_1) \\ &+ (1-\mu_t^1) \, u^2(c_t^2, l_t^2, h_t^2, k_t^1, k_t^2, d_t, n_t, o_t, K_t^2, z_1) \\ &+ \eta_t^C d_t + \eta_t^{NC}(1-d_t) \end{aligned} \tag{3}$$

where $\mu_t^1 \in (0,1)$ denotes the wife's bargaining power within the household. Each spouse $i \in \{1,2\}$ has a distinct utility function $u^i(\cdot)$, reflecting differences in preferences across household members. Throughout, I simply assume that $i = 1$ denotes the wife and

[18] This is a common and standard assumption in the previous literature using dynamic collective models.

[19] The divorce rate in Japan is low—about 1.5 per 1,000—so I assign married couples a low value for the outside option of divorce, such that only a few reach divorce in the model simulations.

$i = 2$ denotes the husband. Despite diverging preferences, the spouses face a joint iid preferences shock toward conception or non-conception, denoted $\eta_t^C d_t$ and $\eta_t^{NC}(1-d_t)$, where the realized shock depends on whether the spouses decide to conceive, indicated by the variable d_t.

The utility of each household member consists of [20]

$$\begin{aligned} u^i(c_t^i, l_t^i, h_t^i, k_t^1, k_t^2, d_t; n_t, o_t, K_t^i, z_1) = & \frac{(c_t^i)^{1+\eta_i}}{1+\eta_i} + \psi_i \frac{(l_t^i)^{1+\gamma_i}}{1+\gamma_i} \\ & + \lambda(n_t, o_t) \frac{(Q_t)^{1+\kappa}}{1+\kappa} + q_i(K_t^i, d_t) \\ & + \mathbf{1}(i=1)\Big[g(z_i, h_t^1) + p(z_i, k_t^1, k_t^2)\Big] \end{aligned} \tag{4}$$

Each spouse derives utility from private consumption c_t^i, private leisure l_t^i, and a home-produced good Q_t, with each component entering utility in the standard constant relative risk aversion (CRRA) form.

The home-produced good captures two components: the welfare of children, if the household has children living at home, and the quality of the household environment, such as cleanliness and homemade meals. The output depends on both spouses' domestic labor supply, k_t^1 and k_t^2, such that $Q_t = Q_t(k_t^1, k_t^2)$. These domestic activities include housekeeping chores—e.g., cleaning and doing laundry—and, if the household has children, childcare duties. The home production technology takes the constant elasticity of substitution (CES) form:[21]

$$Q_t = Q(k_t^1, k_t^2) = \Big(\pi(k_t^1)^\sigma + (1-\pi)(k_t^2)^\sigma\Big)^{\frac{1}{\sigma}} \tag{5}$$

where $\pi \in [0,1]$ captures productivity differences between the spouses' domestic time inputs, and σ controls the degree of substitutability between them.

The weight on home production reflects the importance of the home-produced good for individual utility, and thereby determines the spouses' incentives to allocate time to domestic labor. This weight depends on the number of children in the household, n_t, and the age of the youngest child, o_t:

$$\lambda(n_t, o_t) = B\left[1 + \Big(b_1 \cdot \mathbf{1}(n_t > 0) + b_2 \cdot (n_t - 1)\Big) f(o_t)\right] \tag{6}$$

I assume that spouses' tastes for home production are common within the household

[20] As I do not model the divorce state, I do not include the matching shocks for both spouses in the utility function as previous literature commonly do.

[21] Salazar-Saenz (2022) provide a comprehensive review of commonly used home production specifications. Gayle and Shephard (2019) offer a representative example of the CES specification. Some papers additionally include material inputs in Cobb-Douglas form as a complement to the CES aggregator of time inputs.

and does not differ by gender. The parameter B is the multiplicative scale factor for this taste, and it captures only the quality of living environment when the household has no children. Spouses' additional utility from children is not restricted to be linear in the number of children, allowing the model to capture nonlinearities in how children affect spouses' time use. The parameter b_1 captures the value of being a parent, while b_2 captures the marginal impact of each additional child beyond the first. Apart from the effect of the number of children, this weight decays with the age of the youngest child according to $f(o_t) = \exp(b_3 \cdot o_t)$ as spouses spend less time on childcare as children grow older.

Each spouse has a fixed time budget T^i per period, allocated between market work h_t^i, home production k_t^i, and private leisure l_t^i, so that each individual faces the time constraint

$$h_t^i + k_t^i + l_t^i = T - sleep^i, \quad \forall i \in \{1, 2\} \tag{7}$$

where the time budget $T - sleep^i$ equals 168 hours per week net of sleep. I calibrate weekly sleep time based on gender, number of children, and gender role attitude group (conservative, moderate, or progressive). Childless spouses sleep significantly more than those with children, and spouses in the progressive group sleep less than those in the conservative or moderate groups. Market work and home production hours are directly reported by respondents, while private leisure is measured as the residual.

The fourth term represents the non-monetary, career-related cost associated with having a child in period t. This cost captures the harm of career interruptions—for instance, lower productivity during the fertility period, and reduced promotion opportunities if employers perceive parents as less committed to their careers. The cost differs by gender and is given by

$$q_i(K_t^i, d_t) = \theta_k^i \left(K_t^i + 1\right) d_t \tag{8}$$

where θ_k^i is a cost parameter capturing how costly having a child at time t is for individual i. The cost increases with individual i's human capital stock, K_t^i, since individuals with greater human capital face higher opportunity costs at the fertile period, and therefore bear a larger career-related cost of childbearing.

The fifth and sixth terms apply only to the female spouse and depend on her gender role attitudes, z_i. The fifth term captures her (dis)utility from working long hours. Women with conservative attitudes may be more averse to long work hours, as prevailing social norms tie their sense of duty to household labor and stigmatize women who spend more time on market work. In contrast, women with progressive attitudes may face no such stigma and may even gain utility from working longer hours through a sense of career achievement. The term related to working hours is given by

$$g(z_i, h_t^1) = \psi_{z_i}^{s_t} \left(h_t^1\right)^{\varphi_{z_i}}, \quad i = 1 \tag{9}$$

where $z_i \in \{1, 2, 3\}$ indexes her gender role attitude group, and $s_t \in \{nk, k\}$ indicates whether the couple has children. The scale $\psi_{z_i}^{s_t}$ is specified as a baseline plus an adjustment for having children, $\psi_{z_i}^{k} = \psi_{z_i}^{nk} + \Delta\psi_{z_i}$, where $\psi_{z_i}^{nk}$ is the scale for women without children and $\Delta\psi_{z_i}$ captures how the scale shifts once the couple has children. I do not restrict the sign of either parameter, so whether long hours generate utility or disutility is determined by the estimation. The parameter φ_{z_i} governs the curvature of this (dis)utility and also varies with attitudes.

The sixth term captures the wife's (dis)utility from an unequal division of housework. It depends on the housework gap between spouses and on the wife's gender role attitudes, allowing women to differ in how much they object to, or accept, unequal unpaid work. This term applies only to women, since in Japanese households women typically perform a larger share of family labor, while husbands benefit from home production with limited time investment of their own. Formally, the term is defined as

$$p(z_i, k_t^1, k_t^2) = \theta_{z_i}^{s_t} \max\left(k_t^1 - k_t^2, 0\right)^{\rho_{z_i}}, \quad i = 1 \tag{10}$$

where, as with the preference for working long hours, the scale $\theta_{z_i}^{s_t}$ is specified as a baseline plus an adjustment for having children, $\theta_{z_i}^{k} = \theta_{z_i}^{nk} + \Delta\theta_{z_i}$. Here $\theta_{z_i}^{nk}$ is the scale for women without children and $\Delta\theta_{z_i}$ captures how the scale shifts once the couple has children. The $\max(\cdot, 0)$ operator implies that this term applies only when the wife spends more time on housework than her husband.[22]

The wife's aversion to an unequal division of home production time affects both the household's time allocation and its fertility decisions. This aversion acts as a penalty on the traditional pattern of specialization, thereby leading to a more equal division of housework between spouses. At the same time, an unequal division of domestic labor raises the psychological cost of childcare time, thereby lowering the household's willingness to have children.

Fertility and children. The household decides whether to have another child in period t through a binary choices $d_t \in \{0, 1\}$.

If the spouses decide to conceive, the child is born in the next period with probability of successful conception $\pi(age_t^1)$, which depends on the wife's age. The conception probability accounts for biological fecundity and a potential permanent infertility risk, both calibrated based on the medical literature and detailed in Appendix A.3. I assume no possibility of unplanned fertility, so a new child is not born in the next period if the spouses decides not to conceive.

If the child is successfully born, the number of children living at home is increased

[22] Throughout the model, hours are measured per two-year period. In the two preference terms for working long hours and the housework gap, hours are further divided by 1,000 for numerical stability.

by one unit in the next period, and the age of the youngest child is set to zero. If no childbirth occurs, the number of children remains unchanged, and the age of the youngest child increases by one period unit. Children leave the household according to an empirical move-out rate calibrated from the data.

Human capital and wages. The model allows spouses' wages depend on their human capital stocks as:

$$\ln(w_t^i) = \phi_i^0 + \phi_i^1 K_t^i + \epsilon_t^i,$$
$$\epsilon_t^i \overset{\text{iid}}{\sim} \mathcal{N}(0, s_i)$$

where individual human capital K_t^i is discrete with values $\{0, 1, 2, \cdots, K_{max}\}$ and is accumulated through learning-by-doing. The law of motion for human capital is:

$$K_t^i = \begin{cases} \min\{K_{t-1}^i + 1, K_{\max}\} & \text{with prob. } p_K(h_{t-1}^i)(1 - p_{\delta_i}) \\ K_{t-1}^i & \text{with prob. } p_K(h_{t-1}^i)p_{\delta_i} + \left(1 - p_K(h_{t-1}^i)\right)(1 - p_{\delta_i}) \\ \max\{K_{t-1}^i - 1, 0\} & \text{with prob. } \left(1 - p_K(h_{t-1}^i)\right) p_{\delta_i} \end{cases} \quad (11)$$

The stock increases by one unit with probability $p_K(h_t^i)$, which is increasing in one's working hours at period h_t^i. The functional form for the probability is imposed as $p_K(h_t^i) = 1 - exp(-\alpha_i h_t^i)$, where α_i controls how responsive the human capital process is to changes in work hours. Meanwhile, K_t^i depreciates with exogenous probability p_{δ_i}.

The evolve of human capital allows for accumulation of skills through working experience. If one works longer hours across periods, the human capital is more likely to increase and positively affect wages.

Financial constraint. The household has a budget constraint at each period t:

$$F(c_t^1, c_t^2, n_t) = w_t^1 h_t^1 + w_t^2 h_t^2 + (1 + r)a_t - a_{t+1} \quad (12)$$

where the household expenditure $F(c_t^1, c_t^2, n_t)$ should not exceed the labor income and saving assets.

The household expenditure accommodates child-related expenses and the economies of scale of spouses' private consumption, aligning to Voena (2015); Foerster (2024):

$$F(c_t^1, c_t^2, n_t) = e(n_t)\left((c_t^1)^\delta + (c_t^2)^\delta\right)^{\frac{1}{\delta}}$$

The child-related expenses is a fixed percentage of adult members' private consumption, where the equivalence scale $e(n_t) \geq 1$ increases with the present number of children n_t in the family.

The economies of scale indicates that given fixed expenditure, the spouses are able to consume more than they could if they lived separately. The scaling is shown through an inverse CES function with $\delta \geq 1$.

Bargaining power. Although Lise and Yamada (2019) use the same dataset to show suggestive evidence that Japanese households' decision-making aligns more closely with a limited commitment setting, where wage shocks and marital surplus relative to divorce can shift bargaining weights between spouses, they also present several findings that support the full commitment setup adopted here. First, they find that the arrival of a new child does not significantly alter spousal bargaining power. Second, around 78% of the total variation in spousal bargaining power over the sample period is explained by household heterogeneity at the time of marriage. Among households whose bargaining power is revised over the 20-year sample period, the majority of households (65%) revise the initial weight to a degree of less than 10%. Hence, adopting the full commitment setting does not undermine the paper's primary research question on fertility decisions. Although full commitment may lead to an overestimation of female labor force participation, this framework allows us to focus on the main tradeoff between home production and other aspects of fertility decisions. A limited-commitment extension with an explicit divorce state is left for future work.

Retirement. The model includes a retirement phase for the spouses. Retirement is exogenous at age T_r. There are two key differences in retirement relative to pre-retirement household decision-making. First, both spouses work zero market hours, allocating their time budget between home production and leisure in every period. Second, each spouse receives a pension that is a function of their pre-retirement wage. I assume individuals receive constant retirement income given by $R_s(w_T^i) = 0.388\, w_T^i \overline{h}_i$, where 0.388 is the Japanese net pension replacement rate,[23] w_T^i is the retiree's pre-retirement wage, and $\overline{h}_i$ is the average work hours for females and males, respectively. For simplicity, the model does not allow for divorce in the retirement phase and ignores potential death and bequests.

Recursive Formation. Given the state space, the household makes optimal decisions by maximizing the spouses' weighted utilities over time, and the value function for the household in period t is given by

$$\begin{aligned} V_t^{hh}(\Omega_t) = \max_{\mathbf{C}_t} \quad & \mu_t \Big[u^1(c_t^1, l_t^1, h_t^1, k_t^1, k_t^2, d_t; n_t, o_t, K_t^1, z_1) + \beta \mathbb{E}_t V_{t+1}^1(\Omega_{t+1}) \Big] \\ & + (1-\mu_t) \Big[u^2(c_t^2, l_t^2, h_t^2, k_t^1, k_t^2, d_t; n_t, o_t, K_t^2, z_1) + \beta \mathbb{E}_t V_{t+1}^2(\Omega_{t+1}) \Big] \end{aligned} \tag{13}$$

[23] Source for the net pension replacement rate: OECD (2023).

where β is a time discount factor and $\mathbf{C}_t = (c_t^1, c_t^2, l_t^1, l_t^2, k_t^1, k_t^2, h_t^1, h_t^2, d_t, a_{t+1})$ are the choice variables.

The value function for individual i in period t is shown as:

$$V_t^i(\Omega_t) = u^i(c_t^i, l_t^i, h_t^i, k_t^1, k_t^2, d_t; n_t, o_t, K_t^i, z_1) + \beta \mathbb{E}_t V_{t+1}^i(\Omega_{t+1}), \quad i = 1, 2 \tag{14}$$

The Bellman equation can be decomposed into conditional value functions based on the discrete conception choice made at each period. I index the conditional value functions by C when the household chooses to conceive, and by NC when it chooses not to conceive. Since the conception outcomes are stochastic, I further index the value function by P if the spouses successfully give birth to a new child at the beginning of the following period, and by U otherwise. The conditional value function $V^{hh,U}(.)$ nests two distinct possibilities, households with no conception intention at the previous period and households with a conceive intention but did not result in a successful fertile outcome.

Given the state space, the value of conceiving is written as

$$\begin{aligned} V_t^{hh,C}(\Omega_t) = \quad & \mu_t \, u^1(c_t^1, l_t^1, h_t^1, k_t^1, k_t^2, d_t; n_t, o_t, K_t^1, z_1) \\ & + (1 - \mu_t) \, u^2(c_t^2, l_t^2, h_t^2, k_t^1, k_t^2, d_t; n_t, o_t, K_t^2, z_1) \\ & + \beta \Big[\pi(\text{age}_t^1) \, \mathbb{E}_t V_{t+1}^{hh,P}(\Omega_{t+1}^P) \\ & \qquad + \big(1 - \pi(\text{age}_t^1)\big) \, \mathbb{E}_t V_{t+1}^{hh,U}(\Omega_{t+1}) \Big] \end{aligned} \tag{15}$$

where the first two lines consist of the flow utilities of both spouses. The third line is the future flow of utility if conception is successful, which occurs with probability $\pi(\text{age}_t^1)$. This value depends on the next state space Ω_{t+1}^P, where the superscript P indicates that the number of children increases by one and the age of the youngest child is set to zero. The last line describes the case in which conception is unsuccessful, with continuation value of $V_{t+1}^{hh,U}$, which depends on the next state space Ω_{t+1}.

The value of non-conceiving is given by

$$\begin{aligned} V_t^{hh,NC}(\Omega_t) = \quad & \mu_t \, u^1(c_t^1, l_t^1, h_t^1, k_t^1, k_t^2, d_t; n_t, o_t, K_t^1, z_1) \\ & + (1 - \mu_t) \, u^2(c_t^2, l_t^2, h_t^2, k_t^1, k_t^2, d_t; n_t, o_t, K_t^2, z_1) \\ & + \beta \, \mathbb{E}_t V_{t+1}^{hh,U}(\Omega_{t+1}) \end{aligned} \tag{16}$$

The conception decision is made as

$$V_t^{hh}(\Omega_t) = \max \Big[V_t^{hh,C}(\Omega_t), V_t^{hh,NC}(\Omega_t) \Big] \tag{17}$$

As the preference shocks toward conception and nonconception η_t^C and η_t^{NC}, which are

part of the state vector Ω_t, are drawn from an extreme value distribution, the probability of conception takes a logistic form, with the values of conception and nonconception as arguments:

$$\Pr(d_t = 1 \mid \Omega_t) = \frac{1}{1 + \exp\left(V_t^{hh,NC}(\Omega_t) - V_t^{hh,C}(\Omega_t)\right)} \tag{18}$$

The decision to conceive, noted d_t in Equation (13), is the argmax of expression (17).

The conception choice involves a trade-off between the benefits and costs of having children. Having more children increases the utility from home production for both spouses, but incurs several costs, including career costs, a reduction in private consumption and leisure, and inequality aversion arising from an unbalanced gender role in domestic duties. Conditional on successful conception, spouses spend more time on childcare, especially during the child's early years, which reduces market work hours and private leisure. This reduction in market work further affects household income and wages in two ways. First, shorter work hours lower household income in the current period, reducing household consumption. Second, shorter work hours slow human capital accumulation, thereby affecting future wages. In addition, since female spouses take on the majority of childcare activities after childbirth, domestic duties become more unequal, incurring a higher disutility from unequal domestic duties as a psychological cost for females. Conversely, if the household does not conceive or conception is unsuccessful, spouses enjoy higher consumption and leisure and face lower inequality aversion, but produce less home production.

6 Estimation

The estimation of model parameters proceeds in two steps. First, a small subset of the model parameters is preset to match values from the existing literature and data source. Second, the remaining parameters are estimated directly from the JPSC data using the method of simulated moments (MSM), a numerical technique that estimates parameters by matching moments from simulated data to those from observed data (McFadden, 1989; Gourieroux et al., 1993).

6.1 Parameters set outside the model

The preset model parameters and their values are summarized in Table 2. The annual discount factor and interest rate are set to 0.98 and 0.02, respectively, in line with the literature (Attanasio et al., 2008; Foerster, 2024). The economies of scale in spouses' consumption is set to 1.4, following Voena (2015) and Foerster (2024). The probability of successful conception depends on the wife's age and is calibrated using medical evidence (Trussell and Wilson, 1985; Leridon, 2004; Sommer, 2016), with details provided

in Appendix A.3.

Each period t in the model corresponds to two calendar years, and the total horizon spans $T = 20$ periods, covering women's ages 24 to 64. Accordingly, time-use variables enter the model in hours per period, computed as weekly hours multiplied by 49 working weeks per year and two years per period. Consumption, income, and saving enter in thousand yen per period, obtained by multiplying annual amounts by two. For ease of interpretation, I report all results in weekly hours and monthly thousand yen. Women are assumed to be biologically fertile between ages 24 and 46, during which they face a fertility decision; after age 46, fertility is no longer possible. The model includes two periods of retirement phase, as I assume all agents retire at age 60.[24]

The equivalence scale of child-related expenses is calibrated from the data as a function of the number of children, and differs across the three gender role attitude groups: conservative, moderate, and progressive. This scale captures the additional resources required for children relative to a childless couple, while accounting for economies of scale in spousal consumption. The values defining this child-dependent equivalence scale are detailed in Appendix A.4.

I calibrate the initial distribution of four state variables from the data: the number of children, spouses' human capital stock, and spousal relative bargaining power; further details are provided in Appendix A.5. All women enter the model at age 24, and some already have children in the initial period. I draw each woman's initial number of children and the age of her youngest child from the empirical distribution in the data, conditional on her gender role attitudes. Spouses' initial human capital is calibrated based on education, prior work experience, industry of employment, and region of residence, while the law of motion for human capital is estimated within the model. Spousal relative bargaining power is determined by spouses' relative wages in the initial period, when women are aged 24–25 and their husbands range in age from 20 to 42; I use the wife-to-husband wage ratio as a proxy for bargaining power, capturing spouses' relative economic position at the start of the decision horizon. This relative wage ratio is likewise drawn from the empirical distribution in the data.

The model is solved using a grid search method, in which state and choice variables are discretized and the objective function is evaluated at each grid point to identify the

[24] Japanese employers may set a mandatory retirement age no lower than 60, at which point an employee's original employment contract formally ends. However, employers are required to offer continued employment—typically through a new, often less favorable contract—to any employee who wishes to keep working until age 65. Age 65 is also the age at which workers become eligible for the full public pension; claiming before 65 permanently reduces the monthly benefit by 0.4% for each month claimed early, amounting to a 24% permanent reduction if claimed at 60. Because of this financial disadvantage, most workers do not claim their pension early despite having the option. To ease the computational burden, however, I assume all workers retire and begin claiming their full pension at age 60. I include a retirement phase primarily to provide variation for the backward induction at the end of the model horizon, and since I do not target moments during the retirement phase, this simplification should not affect the estimation results or model fit during the fertile periods.

optimal solution. The main advantage of this approach is that it does not require differentiability of the value function—an important feature here, since the model combines a discrete fertility choice with multiple continuous choices, making differentiability difficult to guarantee. Accordingly, I also preset the grid points for several model variables, listed in Appendix A.4.

Female weekly work hours are discretized into four categories based on the empirical distribution: non-participation (0 hours), part-time (25 hours), full-time (40 hours), and overwork (55 hours). Male weekly work hours are fixed at 60 hours. Annual hours are computed assuming 49 working weeks per year. Each spouse has a weekly time endowment of 168 hours, and sleep time is calibrated from self-reported sleep hours in the data, varying with the number of children and the wife's gender role attitudes. Both the number of children living at home and spouses' human capital stocks are top-coded at 3, to keep the estimation computationally tractable. The age of the youngest child is discretized into six categories: 0–2, 2–4, 4–6, 6–8, 8–10, and 10+.

Table 2: Preset parameters

Parameter	Value	Source
Annual discount factor, β	0.98	Attanasio et al. (2008); Foerster (2024)
Annual interest rate, r	0.02	Attanasio et al. (2008); Foerster (2024)
Economies of scale, δ	1.40	Voena (2015); Foerster (2024)
Probability of successful conception	App. A.3	Trussell and Wilson (1985); Leridon (2004); Sommer (2016)
Child-related expenses, $e(n_t)$		See Appendix A.4, JPSC data
Female and male sleep time		See Appendix A.4, JPSC data
Initial distribution of children		See Appendix A.5, JPSC data
Initial human capital		See Appendix A.5, JPSC data
Initial relative bargaining power		See Appendix A.5, JPSC data

6.2 Estimation

The remaining parameters are estimated by the Method of Simulated Moments (MSM). MSM is commonly used in dynamic household models because the objective function arising from discrete choices is typically non-smooth (Sauer and Taber, 2017). Under this approach, the model is first solved for a candidate parameter vector, and artificial data are simulated from the resulting data-generating process. Moments relevant to the parameters of interest are then computed from the simulated data. The estimation objective is to minimize the distance between the simulated model moments and the corresponding empirical moments from the data:

$$\min_{\theta} \left(m(\theta) - \hat{m}\right)' \widehat{W} \left(m(\theta) - \hat{m}\right).$$

The empirical moments I target are both spouses' weekly home production hours and the wife's weekly work hours, conditional on the number of children, the age of the

youngest child (for households with children), and gender role attitudes. I also target spouses' wage rates conditional on human capital stock, number of children, and the age of the youngest child. As a third set of moments, I target households' average number of children by gender role group by the end of life cycle, to match completed fertility. In total, I target 363 moments. For the weighting matrix $\widehat{W}$, I use a diagonal matrix with the inverse variances of the empirical moments as diagonal entries, adjusting the weights of several moment sets to place additional emphasis on key variables.

All parameters are estimated jointly, but certain moments are especially informative about specific parameters. The wage moments discipline the wage process. Parameters common to all couples are pinned down by the overall levels of time use and how they change with the age of the youngest child. The preferences tied to gender role attitudes, in contrast, are pinned down by differences across attitude groups. Differences in the housework gap across gender role attitudes, and in how these differences change once couples have children, identify the (dis)utility over the housework gap, since neither wages, which are targeted by group, nor parameters common to all couples can generate them. Similarly, differences in wives' market hours across attitude groups, with and without children, identify the preferences over long working hours.

The estimated MSM parameters, along with their asymptotic standard errors (see e.g. Newey and McFadden (1994)), are reported in Table 3. In the wage processes, women's wages start from a much lower level than men's, but rise faster with each additional unit of human capital. Even at the highest level of human capital, however, women's wages remain well below men's. Human capital depreciates with a probability of about 7 to 9 percent per period for both spouses. The probability of accumulating human capital rises with hours worked for both spouses. In home production, the estimated substitutability implies an elasticity of substitution of about 1.2 between spouses' housework, so husbands' time can partly replace wives' time. The weight on the wife's time is about 0.6, indicating that her housework contributes more to the home good than her husband's. The first child substantially raises the value of home production, whereas additional children add little, and this effect fades as the youngest child grows older. The non-pecuniary career cost of fertility rises with each spouse's human capital and is larger for husbands than for wives. As husbands are typically the main earners, this cost likely captures the career pressure a birth places on them, and it may affect the timing of births.

The estimation also shows that the unequal division of labor accompanying children is costly mainly for progressive women, who value market work and object to an unequal share of housework once they become mothers. For the housework gap, the scales are positive for women without children in all three groups. This reflects that childless women perform more housework than their relative wages would predict under an efficient allocation, a deviation absorbed by the preferences tied to gender role attitudes. Having children flips the sign only for progressive women, who become the only group

to experience disutility from the housework gap. For conservative mothers, the utility from a larger gap increases further, while for moderate mothers it decreases but remains positive. The preferences over long working hours point in the same direction. Among women without children, long work hours generate disutility for conservative and moderate women but positive utility for progressive women, consistent with the idea that progressive women derive a sense of career fulfillment from market work. These signs persist after having children, but the magnitudes change, with a slight decline in conservative women's disutility, an increase in moderate women's disutility, and a sharp drop in progressive women's utility. The curvature estimates show that progressive mothers' disutility from the housework gap is nearly linear, so even a small gap is costly to them, while conservative women's aversion to work rises sharply as hours become long.

6.3 Model fit

This section evaluates the model's fit to the data by comparing average outcomes from the model simulations with the corresponding empirical moments. Figure 5 illustrates time allocation for childless spouses. For parents, Figure 6 traces how time use evolves with the youngest child's age. Figure 7 displays the fits for spouses' private consumption, which are untargeted in the estimation. Figure 8 summarizes the fertility patterns.

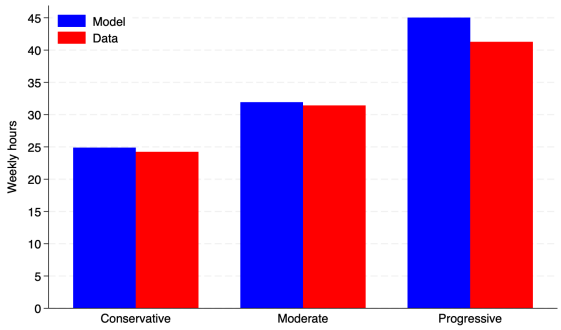
Figure 5 reports average market hours and home production time for childless spouses, most of whom eventually become parents in later years. Each panel plots model prediction and empirical means across the three gender-view groups. Panel (a) reports females' market hours; since husbands' market hours are fixed at 60 per week, their fit is neither targeted nor displayed. Home production hours appear separately for females in panel (b) and husbands in panel (c). [25] The simulated moments track the data closely, and the model reproduces cross-group variation well. Relative to conservative women, moderate women work 29% more market hours and progressive women 71% more. In contrast, female home production hours vary little across groups in levels. Moderate women's share exceeds conservative women's by 5%, while progressive women's falls short by the same margin. Husbands show negligible absolute differences in home production but sizable percentage differences, with husbands of moderate females contributing 23% more than husbands with conservative wives, and husbands who have progressive wives work 78% more at home.

For spouses who have children, Figure 6 shows the trend of females average market work across age of the youngest child in Plot (a), and home production hours for females in Plot (b) and for their husbands in Plot (c). The figure pools across gender-view groups, so it captures the population-wide pattern. For group-specific behavior, the figures in

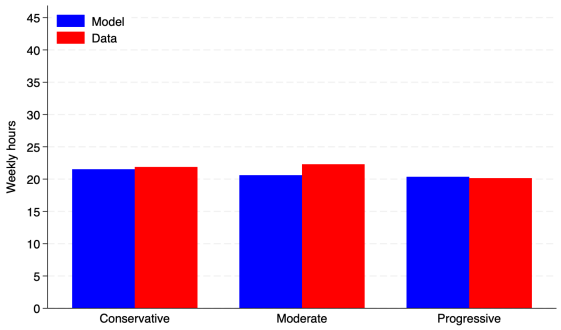
[25] The outcomes for childless families reflect the cross-sectional mean across agents when they are childless and therefore differ from the event-study estimates in 3, which are computed across event time.

Table 3: Estimated parameters

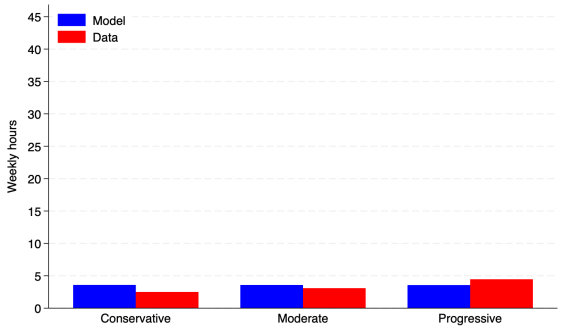
Parameter	Estimate	Standard error	Description
Leisure preferences			
γ_1	-0.6995	0.0120	Curvature, female
ψ_1	0.5364	0.0394	Scale, female
γ_2	-0.3158	0.0094	Curvature, male
ψ_2	0.0931	0.0046	Scale, male
Wage processes			
ϕ_1^0	-0.9445	0.0342	Intercept, female
ϕ_1^1	0.2597	0.0114	Return to human capital, female
s_1	0.9089	0.0299	Residual s.d., female
α_1	0.0003	1.35×10^{-5}	Human capital accumulation, female
p_{δ_1}	0.0686	0.0095	Human capital depreciation, female
ϕ_2^0	0.0665	0.0111	Intercept, male
ϕ_2^1	0.2077	0.0143	Return to human capital, male
s_2	0.5223	0.0248	Residual s.d., male
α_2	0.0001	5.17×10^{-6}	Human capital accumulation, male
p_{δ_2}	0.0933	0.0339	Human capital depreciation, male
Home production			
κ	-0.8427	0.0036	Curvature
B	1.1188	0.0393	Scale
σ	0.1651	0.0023	Substitutability between spouses' time
π	0.5958	0.0218	Weight on wife's time
b_1	4.4774	0.1454	First child value
b_2	0.0070	0.0052	Additional child value
b_3	-0.2038	0.0043	Decay with youngest child's age
Non-monetary career cost of fertility			
θ_k^1	-1.1585	0.4316	Cost, female
θ_k^2	-1.7546	0.6569	Cost, male
(Dis)utility from housework gap			
$\theta_{z_1}^{nk}$	0.0114	0.0026	Scale (no child), conservative
$\theta_{z_2}^{nk}$	0.0574	0.0086	Scale (no child), moderate
$\theta_{z_3}^{nk}$	0.2716	0.0469	Scale (no child), progressive
$\Delta\theta_{z_1}$	0.0436	0.0091	Child adjustment, conservative
$\Delta\theta_{z_2}$	-0.0471	0.0061	Child adjustment, moderate
$\Delta\theta_{z_3}$	-0.4641	0.0844	Child adjustment, progressive
ρ_{z_1}	2.0000	0.0724	Curvature, conservative
ρ_{z_2}	1.9890	0.1641	Curvature, moderate
ρ_{z_3}	1.1309	0.1150	Curvature, progressive
(Dis)utility from working long hours			
$\psi_{z_1}^{nk}$	-0.5743	0.0792	Scale (no child), conservative
$\psi_{z_2}^{nk}$	-0.3289	0.0421	Scale (no child), moderate
$\psi_{z_3}^{nk}$	0.6091	0.0822	Scale (no child), progressive
$\Delta\psi_{z_1}$	0.0186	0.0048	Child adjustment, conservative
$\Delta\psi_{z_2}$	-0.1005	0.0236	Child adjustment, moderate
$\Delta\psi_{z_3}$	-0.4904	0.0651	Child adjustment, progressive
φ_{z_1}	1.5902	0.1063	Curvature, conservative
φ_{z_2}	1.0399	0.1257	Curvature, moderate
φ_{z_3}	1.0056	0.0966	Curvature, progressive



(a) Females' market work hours



(b) Females' home production hours



(c) Males' home production hours

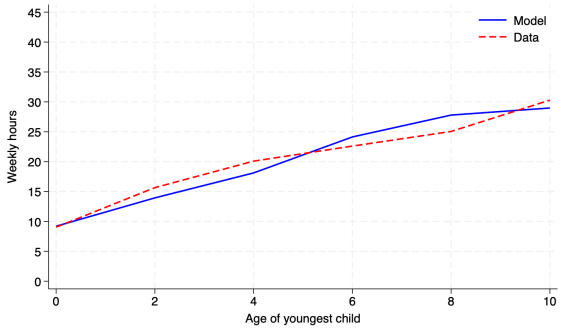
Figure 5: Childless families: market work and home production

Appendix breakdowns separately, with market hours by attitude group in Figure A10, females' home production in Figure A11, and husbands' home production in Figure A12.

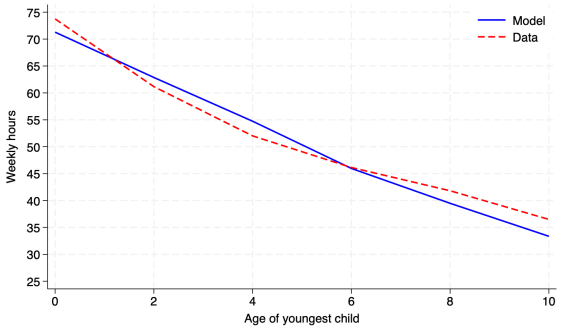
In general, the model predicts the trend well. Females increase their market work hours and reduce home production hours as the child grows, and husbands' housework time follows the same declining pattern. Due to grid discretization, child age is grouped into 0–2, 2–4, 4–6, 6–8, 8–10, and 10–18, so the value plotted at age 10 reflects outcomes for youngest children aged 10 to 18. Fit quality varies in the breakdown of gender-role groups. Female market work is predicted relatively well for conservative and moderate group, but the model overpredicts progressive women's hours once the child reaches age 4 and above. A plausible explanation is that progressive women return to the labor market sooner than other groups and reach a plateau in hours earlier. Those who intend to return do so in the early years, so the growth rate does not persist later on in data, whereas the model imposes an identical housework decay rate across ages and cannot generate this flattening.

In addition, the model underpredicts the home production hours of conservative and progressive women once the youngest child reaches 6 years old and above. This likely stems from two reasons. First, although the model matches the completed fertility rate well, it overstates the early birth rate when females are young, for reasons I outline in the subsequent paragraphs. In the data, however, progressive women are more likely to postpone childbirth, but the model fails to capture this. Children in the simulation are older

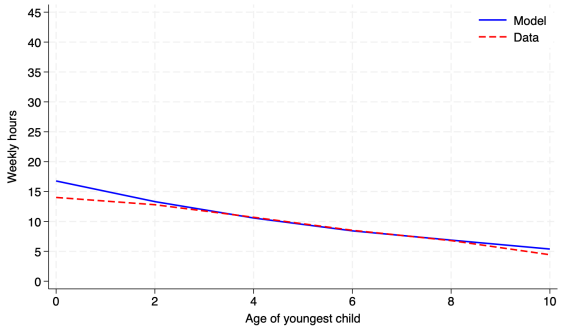
than the data reports, which accelerates the decaying rate of home production hours for both spouses. Second, the model calibrates children's move-out rate according to child age, based on the empirical distribution, and sets this rate identical for all groups. In reality, some children may still live with their parents, leading to an model under-prediction of domestic labor investment in later years. This mainly affects the conservative group, since the data show the move-out rate of adult children for conservative women is lower than for other groups, particularly for the first child.



(a) Females' market work hours



(b) Females' home production hours



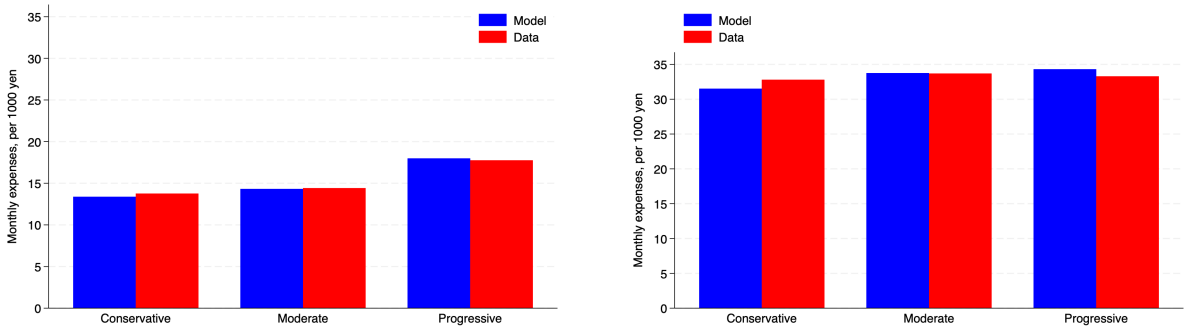
(c) Males' home production hours

Figure 6: Market work and home production across age of youngest child

The model also reports the fit of untargeted moments. One of these is the private consumption of each spouse, shown in Figure 7, with females' average consumption in panel (a) and husbands' average consumption in panel (b). I present agents' average expenses across gender-view groups in separate bars. As a supplementary check, the fit for childless couples and across the age of the youngest child are displayed in Figure A13.

Spouses' consumption rises steadily with the progressiveness of their gender-role attitudes. The first reason is the income effect. Progressive women are more likely to participate actively in the labor market, with women's labor participation rates of 41.7%, 55.6%, and 69.4% across the three groups, and longer average working hours for more progressive women as shown in the earlier patterns. Meanwhile, they earn higher wages and thus have greater bargaining power, proxied by the relative wage rate between spouses; this fit is shown in Figure A14. Due to the bargaining effect, progressive women gain a higher consumption share than other groups in the intrahousehold expense allocation.

Meanwhile, male consumption rises with their wives' gender-role attitudes due to the income effect from assortative matching.



(a) Females average consumption **(b)** Males average consumption

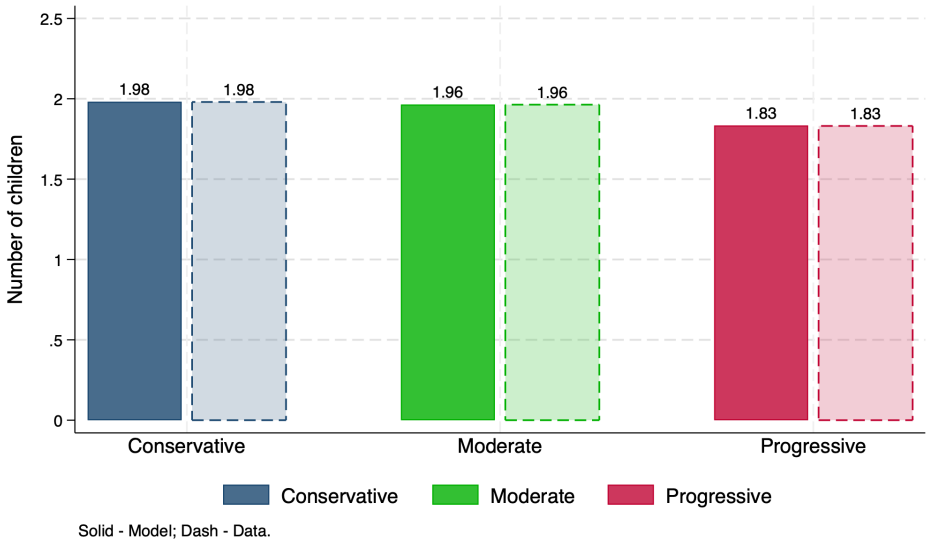
Figure 7: Untargeted moments: consumption

Figure 8 presents the model fit for fertility outcomes. Panel (a) shows completed fertility, defined as the average number of children per household when the wife is 46. Panel (b) shows the average number of children by female age; although this moment is not targeted, the model fits it relatively well.

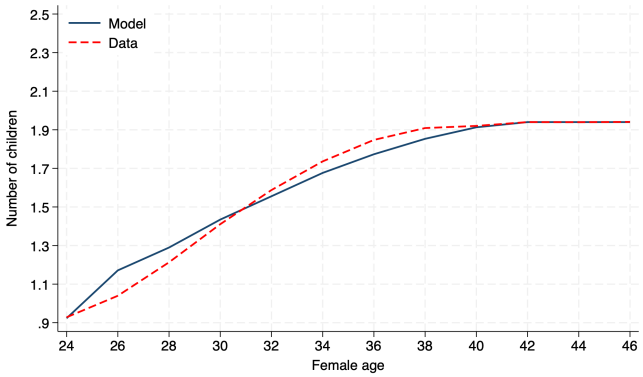
Panel (a) shows a good fit for the number of children across gender-view groups. Panel (b) shows the average number of children across female age, reflecting the cumulative birth rate of the sample, with the x-axis representing age bins across time periods. The model predicts an earlier birth rate than observed in the data. This is because the model captures three factors in the birth decision: child-related expenses, career costs, and personal beliefs about domestic labor. In reality, however, couples may also weigh other considerations, such as housing space, child-free lifestyle, and physical conditions, generating further postponement in childbirth timing. Nevertheless, the model converges to the data moments after female age 36, providing a reliable pattern for analyzing completed fertility.

Some samples do not report the arrival of children that occurred before respondents entered the survey at age 24. As a result, an initial number of children is assigned to females by gender-view group, matched to the empirical distribution. Figure A15 presents the composition of the average number of children assigned to each gender-view group at the initial period and the subsequent increase predicted by the model. Conservative females start with an average of 0.92 children, moderate females with 0.95, and progressive females with 0.88. The three groups subsequently increase their average number of children by 1.07, 1.01, and 0.95, respectively. This pattern shows that the higher fertility outcome among the conservative group is not driven by initial children, but by a higher birth rate that emerges endogenously from the mechanisms incorporated in the model.

Overall, the simulated moments align closely with the empirical moments at the mean



(a) Completed fertility



(b) Untargeted Moment: Number of children by age

Figure 8: Fertility pattern

level. The model's prediction of consumption should be interpreted with caution, but the predictions for time use largely match the data moments and exhibit a good overall fit.

7 Counterfactual Analyses

This section uses the estimated parameters to examine, through counterfactual simulations, the channels that shape couples' fertility decisions and intra-household allocation. I first assess the role of gender role attitudes in these outcomes. I then examine how a reduction in husbands' market work affects fertility and time allocation, and whether a more equal division of labor between spouses raises fertility further.

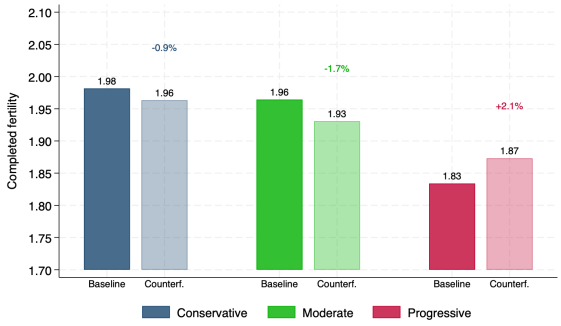
7.1 Gender view channel

I assess the role of the gender view channel using two counterfactual exercises. I first examine whether, and in which direction, the channel affects fertility decisions and time allocation. I then quantify how much of the fertility differences between groups it explains, which also indicates how fertility may evolve if attitudes continue to shift in a progressive direction over time.

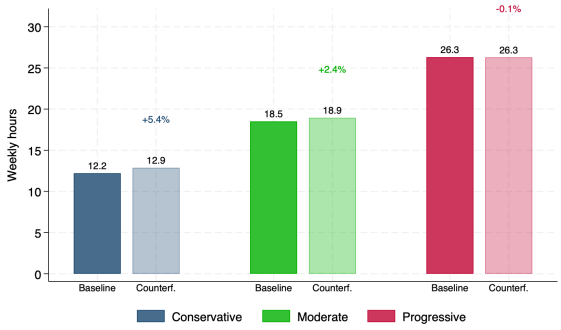
In the first simulation, I shut down the gender view channel by eliminating women's perceived unfairness of housework inequality and their (dis)utility from working long hours (Equation 9 and Equation 10). Figure 9 reports the resulting completed fertility and time allocation of women. The results suggest that gender role attitudes reinforce household specialization and raise fertility among conservative and moderate women, while discouraging domestic labor and lowering fertility among progressive women. Without the preferences associated with gender role attitudes, completed fertility falls by about 1% among conservative women and 1.7% among moderate women, but rises by 2.1% among progressive women. The adjustment in time allocation mirrors this asymmetry and operates predominantly through unpaid work. Conservative women reduce their housework by 4.6 hours per week (9%) while increasing their market work by only 0.7 hours. Moderate women adjust in the same direction on a smaller scale, whereas progressive women slightly increase their housework and reduce their market work. These patterns point to the mechanism behind the fertility responses, in which traditional attitudes lower the non-pecuniary cost of the unequal division of labor that accompanies children, whereas progressive attitudes raise it.

Moreover, the fertility responses contrast with the changes in time allocation. Gender views distort the time allocation of conservative and moderate women the most, yet their fertility responds only modestly. Progressive women display the opposite pattern, with the smallest change in time allocation but the largest fertility response. The non-pecuniary costs thus appear to weigh more heavily on the fertility decisions of progressive

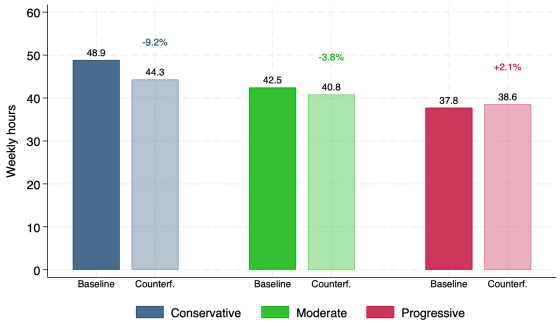
women than on their time use. While these results establish that gender role attitudes affect fertility, they do not reveal how much of the fertility gap between groups the attitudes explain. The second simulation addresses this question.



(a) Completed fertility



(b) Women's market work



(c) Women's home production

Figure 9: Outcomes without the gender view channel

Gender role attitudes are not the only factor that shapes fertility. Women with different attitudes also differ in education, earning potential, and the economic constraints they face, so the observed fertility gap between groups reflects both their views and other characteristics. How much of the gap gender role attitudes account for remains an open question, and the answer matters for policy in two ways. First, it identifies women whose fertility is held back by economic constraints, for whom policies such as childcare subsidies are more promising. Second, since attitudes are steadily shifting toward progressive views, policymakers need to know how much fertility would fall if more women held progressive views, and whether this decline operates through women's market work or the division of childcare. Quantifying this decline is particularly relevant while gender roles are in transition, since it helps assess whether policies that make the division of labor within households more equal can offset it.

The second exercise assigns progressive views to all women while holding their other characteristics fixed. Specifically, I set the gender-view-specific parameters of conservative and moderate women equal to those of progressive women, including women's utility or disutility from market work and their perceived unfairness of an unequal domestic labor. Conservative and moderate women thus face a lower disutility of market work and

a stronger preference for an equal division of housework. All other characteristics, including spouses' initial human capital and their preferences over intra-household allocation, remain unchanged.

Figure 10 compares the resulting completed fertility with the baseline. Conservative women, whose views differ most from those of progressive women, reduce their completed fertility by only 2.5% (0.05 children), whereas moderate women reduce theirs by 4.0% (0.08 children). The shift in views therefore closes 33.7% of the fertility gap between conservative and progressive women and 60.2% of the gap between moderate and progressive women. The gender view channel thus accounts for a substantial share of the fertility differences across groups, particularly for moderate women. The remaining gap reflects other differences across groups, such as earning potential, which matter more for conservative women, given the smaller share of their gap explained by gender views.

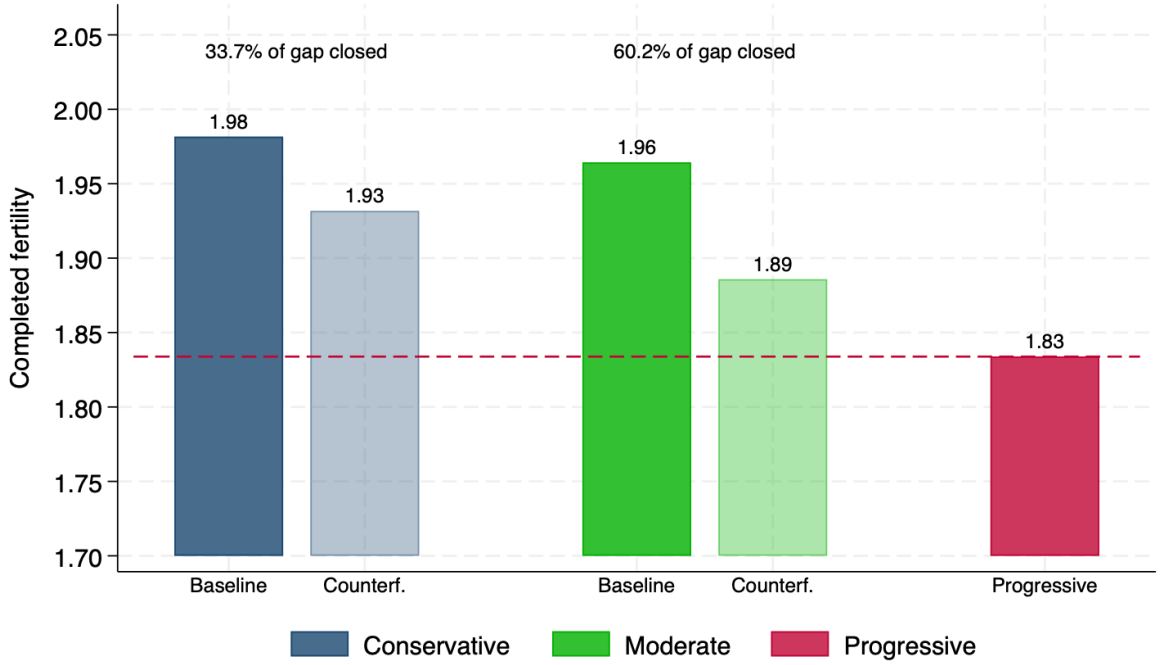


Figure 10: Completed fertility if all women hold progressive views

The results are accompanied by several changes of households allocation following the gender view change shown in Figure A16. As expected, conservative and moderate women's market work increases and home production time reduces. Driven by the human capital accummulation, women's hourly wage rate increases. Since husbands market work is fixed at 60 hours per week, the spouses overall household incomes increases, which leads to the increase of private consumption for both spouses. Besides, husband's home production hours slightly reduces due to the declining fertility outcomes. This pattern suggests that the gender view channel mainly change women's paid and unpaid work pattern, despite the husbands behaviors barely change, the spouses relative labor

division is prone to be more equal. This increase the non-pecuniary costs of deviating from such equal division and having children. The change also increases the pecuniary costs, by both spouses higher personal expenses.

I further disentangle the effects by examining the responses of women with varying initial human capital but the same gender role attitudes, displayed in Figure 11, with absolute levels shown in Figure A17. I measure human capital at the initial period to isolate its effect before shifts in gender role attitudes alter housework and market-work time allocation. Initial human capital reflects education and past work experience at female age 24, and is a good proxy for potential wages around the time of marriage and childbirth. By equipping them with progressive views, previously conservative women with varying human capital backgrounds reduce their completed fertility by 1.8-4.3 percentage points, corresponding to a reduction in the number of children of around 0.04-0.08. Conservative women with higher initial human capital, likely those with higher potential wages, show a larger decrease in their number of children. The pattern for the moderate group differs. Completed fertility declines at every level of female human capital, and the decline becomes smaller as initial human capital rises from 0 to 2, falling from 4.23% to 3.72% and then to 2.56%. Women with the highest initial human capital, however, show the largest decrease, reducing their completed fertility by 4.88 percentage points.

To understand these fertility patterns, I examine how women's private consumption and market work respond by initial human capital within each gender-view group in Figure A17. Women in both groups substantially increase their market work. Conservative women roughly double their weekly hours, with increases ranging from 13.5 hours for women with the lowest human capital to 11.9 hours for those with the highest. Moderate women, who already work more in the baseline, raise their hours by a smaller 6.2 to 7.1 hours per week. These increases in labor supply translate into higher private consumption, by 2.1 to 2.4 thousand yen for conservative women and 1.1 to 1.4 thousand yen for moderate women. Adopting progressive views thus leads women to devote more time to market work, raising their earnings and consumption. As a result, the relative value of additional children declines, and completed fertility falls in every group.

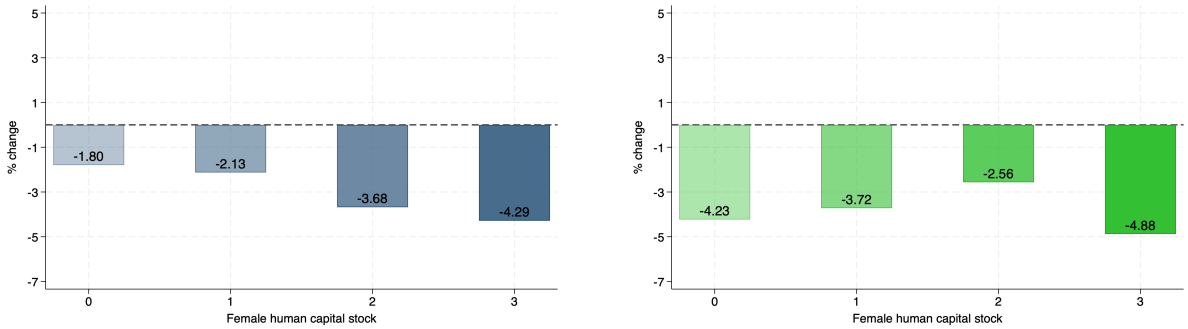
The size of the fertility decline varies with women's initial human capital, but the pattern differs between the two groups. Two forces shape the fertility response. The first is the opportunity cost of children. With progressive views, women devote more time to market work and earn more, so time spent on children becomes more costly. The second is a more balanced division of housework. Because women's more progressive views shift the intra-household allocation toward greater equality,[26] the housework gap between spouses narrows, which reduces the penalty from unequal gender roles for mothers and makes additional children relatively less costly. The first force dominates among conservative

[26]The counterfactual assumes that husbands' preferences are unchanged, so the more equal allocation is driven by women's more progressive views.

women, while the second plays a larger role among moderate women. Among conservative women, the decline grows with initial human capital, from 0.04 fewer children at the lowest level to 0.08 at the highest. Conservative women work few hours in the baseline, so all of them increase their market work by a similarly large amount, and differences in wages determine the fertility response. Women with more human capital earn higher wages, which makes market work more rewarding and children relatively more costly. Among moderate women, the decline instead shrinks as initial human capital rises from level 0 to level 2, from 0.08 to 0.05 fewer children. Since moderate women already work more in the baseline, the additional earnings from working more are smaller than for conservative women, and the more balanced division of housework makes additional children less psychologically costly. Women with the highest human capital break this pattern and show the largest decline, with 0.10 fewer children, as their high earnings potential makes the opportunity cost of children dominate.

Overall, the counterfactual analysis reveals that gender views channel accounts for 60.2% of the fertility premium between moderate and progressive women, and 33.7% between conservative and progressive women. This aggregate effect, however, masks considerable heterogeneity by female's initial human capital, which relates to their educational attainment and earning potential. Among conservative women, the fertility decline grows with initial human capital, as women with better career prospects face a higher opportunity cost of children once they adopt progressive views and expand their market work. Among moderate women, the decline shrinks as human capital rises from the lowest level to the second highest, as these women already work more in the baseline and the more equal labor division under progressive views partly offsets the decline. Across both groups, the women with the strongest earnings potential are therefore the most sensitive to the costs that progressive views place on motherhood.

These findings have several policy implications. First, more progressive gender role attitudes alone reduce fertility in every group, as women shift their time toward market work and the opportunity cost of children rises. Progress toward egalitarian attitudes therefore needs to be accompanied by policies that make it easier to combine work and motherhood. Second, the largest declines occur among women with the highest earnings potential, 0.08 fewer children among conservative women and 0.10 among moderate women, for whom the opportunity cost of children is greatest. Policies that lower the career cost of motherhood, such as accessible childcare, job-protected parental leave, and support for returning to work after birth, are particularly relevant for these women. Third, a more equal division of housework between spouses mitigates the fertility decline. Policies that increase men's involvement in home production, such as paternity leave or limits on long working hours, can therefore complement changes in gender norms and help sustain fertility as attitudes become more egalitarian.



(a) Conservative **(b)** Moderate

Note: The framed segment in Panel b shows the simulated increment from the counterfactual, relative to the baseline value shown in the unframed segment below.

Figure 11: Changes in completed fertility by human capital stock (%)

7.2 Husband's overtime work and domestic labor

Despite the legal limit of 40 working hours per week, men's actual work hours show little variation, remaining around 60 hours per week even around the time of childbirth. This persistence reflects workplace norms that treat long hours as a signal of dedication, pressuring employees to stay late regardless of formal regulations. The model captures this feature of the work environment by fixing men's working time at 60 hours per week, leaving husbands only about 33 hours per week for home production and leisure—a constraint that substantially intensifies the unequal division of childcare between spouses.

I counterfactually reduce men's market work hours from the current 60 to the legal limit of 40 hours per week and examine the resulting fertility responses and reallocation of freed-up time. Figure 12 reports completed fertility across gender role groups, with the percentage change relative to baseline reported in Figure A18.

Reducing husbands' weekly market work raises completed fertility in all three groups. Husbands' long working hours appear to act as a binding constraint on fertility, as freeing up their time allows couples to have more children. I first consider a modest reduction from 60 to 54 hours per week, which corresponds to a 10% cut in husbands' working time and represents a realistic policy target for husbands working long hours. Under this scenario, progressive women show the largest response, as their completed fertility rises by 0.08 children to 1.92 and lies 4.4% above the baseline. The effect is somewhat smaller for conservative women, whose completed fertility increases by 3.7% to 2.06 children. Moderate women respond the least. For this group, completed fertility rises by only 0.03 children to 1.99, which amounts to an increase of 1.3%. We then consider a larger reduction to the statutory 40-hour workweek, which amplifies these gains substantially. Progressive women again respond most strongly, as their completed fertility rises by 16.6% to 2.14 children. Among conservative women, completed fertility reaches 2.25 children and is thus 13.5% higher than in the baseline. The increase is smallest for moderate

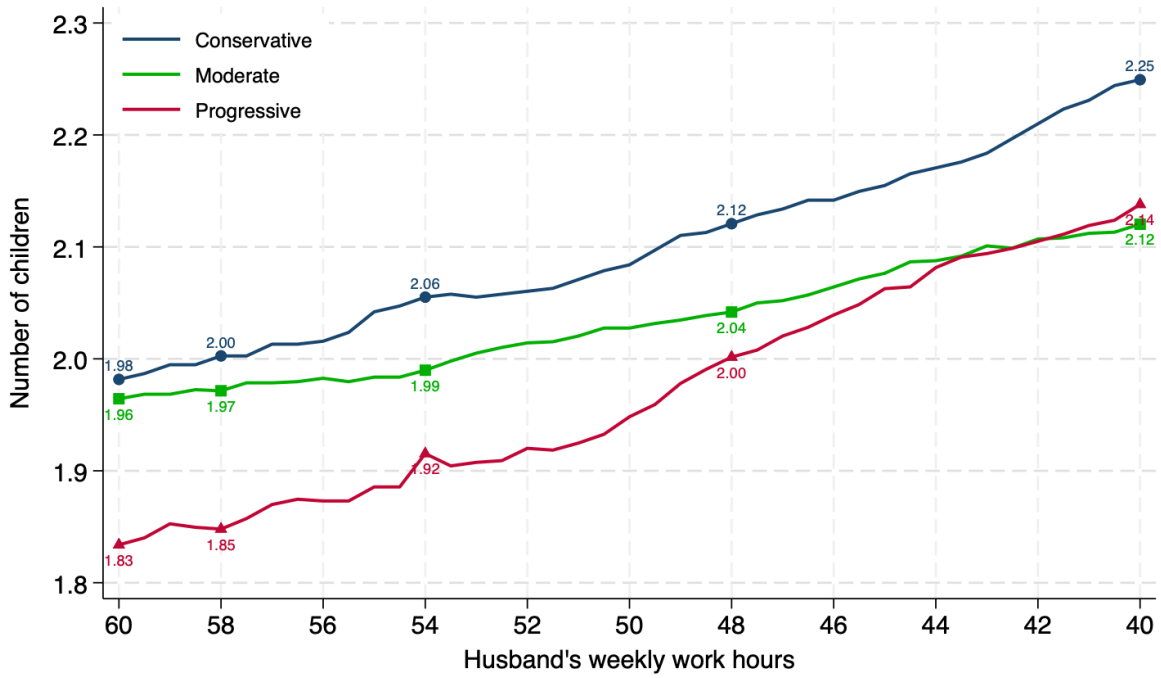
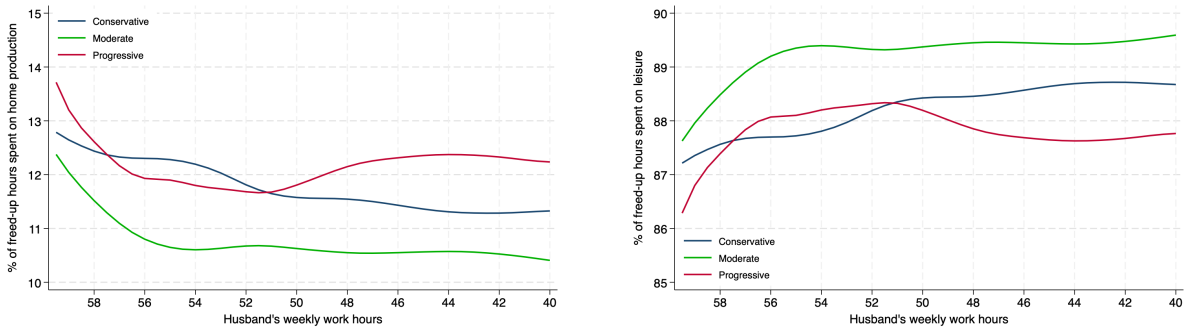


Figure 12: Completed fertility if husbands reduce market work

women, whose completed fertility rises by 7.9% to 2.12 children. In both scenarios, progressive women respond most to shorter working hours, followed by conservative and moderate women, since progressive women devote the most time to market work and bear the highest non-pecuniary costs of an unequal division of household labor. Additional paternal involvement at home therefore reduces their opportunity cost of childrearing the most, which gives them the strongest incentive to have more children.

The fertility effects are closely linked how husbands reallocate the time released from market work. Figure 13 decomposes this freed-up time into home production and private leisure. The reallocation is strongly tilted toward leisure, with husbands devoting 86-90% of the released time to private leisure and only 10-14% to home production. Male home production therefore increases only modestly, from below 9 hours per week in the baseline to between 11 and 12 hours per week (see Figure A19 in the Appendix for the absolute changes). The level of paternal involvement varies systematically with gender role attitudes. Husbands of conservative women contribute the most to home production as they have more children, while husbands of moderate women contribute the least. Among progressive group, however, husbands' home production rises markedly once husbands' market hours fall below 50 per week, coinciding with the sharper increase in fertility in this group and the correspondingly higher demand for childcare.

Women's time allocation and wages adjust in response to the reduction in husbands' market work (Figure A19 in the Appendix). Shorter working hours for husbands induce women to expand their own market work and reduce their private leisure, consistent with



(a) Reallocated hours: home production (%) **(b)** Reallocated hours: leisure (%)

Figure 13: Husbands' allocation of freed hours (%)

an added-worker effect that partly compensates for the decline in household earnings. The magnitude of this response, however, depends on women's initial attachment to the labor market. Progressive women supply the most market work in all scenarios, yet their relative adjustment is the smallest given their already high baseline hours. When husbands' hours fall from 60 to 40, progressive women increase their market work by less than two hours, from 26.3 to 28.1 hours per week. Conservative and moderate women start from considerably lower levels and therefore exhibit larger relative gains of roughly 20% and 14%, which correspond to 2.5 and 2.7 additional hours per week, respectively. Women's home production, in contrast, responds only marginally, indicating that it remains sticky despite husbands' greater involvement in childcare. Conservative and moderate women barely change their home production hours. Progressive women follow a different pattern. Their home production increases slightly once husbands' hours fall to 50 and rises roughly twice as fast below that threshold, reflecting the stronger fertility response and the resulting higher demand for childcare in their households. Wages adjust in line with these changes in market work. Wives' hourly wages rise through human capital accumulation, while husbands' wages decline due to their reduced market work. When husbands' hours fall from 60 to 40, conservative wives' wages increase by about 4.6% and their husbands' wages decrease by about 3%. Among moderate couples, the two changes are of similar size, at 2.9% and 2.8%, whereas progressive wives' wages rise by only 1%, against a 3% decline for their husbands. Household earnings nonetheless fall substantially in all groups. Husbands earn considerably more and lose a third of their working hours, a loss that wives' modest increase in market work cannot offset. This decline in household earnings is a key challenge for households when husbands reduce their working hours.

Japan's 2018 Work Style Reform was introduced within a broader policy agenda to counter declining fertility and low female labor force participation, and it established, for the first time, legally binding caps on overtime beyond the statutory 40-hour workweek. Evaluations of the reform find a 25% decline in average monthly overtime, roughly three

fewer weekly hours for men above the cap, and slightly higher hourly wages (Burdin et al., 2026).[27] Taking this reduction as an exogenous shift in husbands' working time, I simulate the reform's implications for fertility. The model predicts an increase in completed fertility of up to 0.036 children (2%) in the progressive group, 0.031 children (about 1.6%) in the conservative group, and 0.014 children (0.8%) in the moderate group. These magnitudes are likely to understate the reform's effect slightly. The simulation treats the reform as a shock to husbands' hours alone, although women working overtime also reduced their hours. Shorter hours for wives would free more time within the household, which could further raise fertility, but would also reduce their earnings, which works in the opposite direction.[28] Because only 9.3% of wives in my sample work more than the cap allows, however, this bias is likely small.

These predictions cannot yet be compared with direct evidence, as no study has estimated the reform's effect on fertility and the decline in fertility since 2019 is confounded by the COVID-19 pandemic. Identifying this effect would require data that link both spouses' working hours to fertility outcomes, a task I leave for future research. Nevertheless, the available evidence is consistent with the model's predictions. Husbands spend 11–15% of the freed time on domestic work in the model, less than one hour per week for the three hours work reduction, which is consistent with the absence of a significant change in housework time in Burdin et al. (2026). They also find, as suggestive evidence, that married women where both spouses were treated by the reform report higher satisfaction with their spouse, in line with the non-pecuniary cost of an unequal division of labor in my model. In addition, Nagase and Brinton (2017) finds that Japan's mandated short-hours option raised first births among childless working women, suggesting that working-time policies can affect fertility. The fertility predictions should therefore be read as the model's answer to a question that the data cannot yet address.

I next examine whether fertility rises further when husbands devote a larger share of their freed-up time to home production. In this counterfactual, husbands devote half of freed-up time to housework, and women re-optimize their time allocation. The scenario therefore represents a more equal division of labor between spouses. For fertility decisions, this reallocation involves a trade-off. On the one hand, departing from household

[27] Market work hours in this paper are not directly comparable with those in Burdin et al. (2026). In their survey data, men work about 53 hours per week, compared with about 60 hours for husbands in my sample. Three differences explain most of the gap. First, market work in this paper includes commuting time, about 6 hours per week for husbands and 2 hours for wives, whereas Burdin et al. (2026) measure working hours only. Second, their sample is restricted to full-time, non-managerial employees aged 20–65, whereas my sample of husbands also includes managers and self-employed men, who are exempt from the overtime cap and often work longer hours. Third, husbands in my sample are married, mostly prime-age, and often fathers, the group with the longest working hours. Because the cap does not affect commuting time, I apply their estimated reduction in weekly working hours to husbands' hours directly, rather than as a proportion of total hours.

[28] Using payroll data, Burdin et al. (2026) find that total earnings fell by only about 1.4%, suggesting that the time effect is likely to dominate.

specialization lowers household earnings, since women earn lower wages on average, and women's greater labor market attachment raises the career costs of childbearing through human capital accumulation. On the other hand, a more equal division of labor reduces the non-pecuniary costs that women bear under an unequal allocation of domestic work, particularly among progressive women. Whether fertility increases further is therefore an open question.

Figure A20 compares fertility and other outcomes across two scenarios. In the first, spouses re-optimize their time allocation in response to husbands' shorter market hours. In the second, husbands spend half of their freed-up time on home production, which replaces part of their wives' home production, and wives re-allocate the time released between leisure and market work. Relative to the first scenario, the more equal division of labor further raises fertility. At 40 working hours, completed fertility increases by an additional 1.7 percentage points among moderate and progressive women and 1.5 percentage points among conservative women, corresponding to about 0.03 additional children in each group.

Husbands' additional home production substitutes only partly for wives'. About 60% of husbands' additional home production replaces wives' home production, while the other 40% comes on top and raises the couple's total home production, reflecting both the larger number of children and the persistence of wives' domestic work. When husbands' hours fall to 40, for example, every 10 additional hours of husbands' home production reduce wives' home production by about 6 hours. Wives allocate the time released between leisure and market work, with about one third going to additional market work. Relative to the baseline, conservative women's market work therefore rises by 32% to 18.3 hours per week, compared with 20% in the first scenario. The corresponding increases are 22% to 25.6 hours for moderate women, compared with 14%, and 14% to 33.9 hours for progressive women, compared with 7%.

Taken together, the results point to two positive outcomes. First, a more equal division of labor raises fertility beyond the gains from shorter working hours alone, which shows that the division of labor matters for fertility decisions. Second, combining shorter working hours for husbands with a more equal division of labor expands female labor supply without compromising fertility, while also giving men better work life balance. Such a policy mix could help narrow gender gaps in the labor market and weaken the culture of long working hours in Japan.

The additional fertility gain from the reallocation is, however, considerably smaller than the gain from husbands' shorter work hours itself. Two factors explain this. First, even in the first scenario, husbands devote 11–15% of their freed-up time, or about 2–3 hours per week, to home production, which already narrows the gender gap in domestic work. Part of the fertility gain from a more equal division is therefore already captured by the first scenario. Second, compared with the first scenario, the reallocation entails

additional costs that offset part of its benefits for fertility. In the first scenario, husbands' additional non-market time raises their utility through both more leisure and more home production. The reallocation, however, shifts part of non-market time from leisure to home production, raising the childcare time husbands must invest. This may lower husbands' birth intention. In addition, wives' additional market work further raises their wages through human capital accumulation, which increases the opportunity cost of children. Nevertheless, fertility still rises above its level in the first scenario, implying that the benefits of a more equal division of labor for wives outweigh these costs.

8 Conclusion

Fertility has fallen as women's opportunities have expanded faster than the division of labor within households has changed. This paper studies how the resulting non-pecuniary costs depend on women's own gender role attitudes. The paper uses data from the Japanese Panel Survey of Consumers, and estimates a life-cycle model in which spouses jointly determine their consumption, time use and fertility.

Three results stand out. First, an unequal division of labor carries a non-pecuniary cost beyond forgone earnings, which falls mainly on progressive women. The costs women attach to an unequal division account for 60% of the fertility gap between moderate and progressive women and a third of the gap between conservative and progressive women. This cost arises from the interaction between women's gender views and the prevailing division of paid and unpaid work. Second, husbands' long working hours are a binding constraint on fertility, and a little paternal time goes a long way. Cutting husbands' hours by 10%, from 60 to 54 a week, raises completed fertility by 4.4% among progressive women, 3.7% among conservative women, and 1.3% among moderate women, even though husbands devote only 10–14% of the freed time, less than one hour per week, to domestic labor. Third, when husbands' shorter hours are accompanied by a more equal division of housework, women's labor supply rises further and completed fertility also rises slightly.

These results matter beyond Japan. In South Korea and other countries where long working hours and unequal gender roles coexist with rapidly changing attitudes, family policies that target only mothers, such as maternity leave and childcare support, address only part of the problem. Limits on long working hours, usually discussed as labor market policy, are also family policies that shape fertility and narrow the gender gap in the division of labor at home.

The framework opens several avenues for future research. First, this paper focuses on women's gender role attitudes, yet husbands' attitudes may also interact with their wives' and jointly shape couples' fertility and allocation decisions, a question that builds on evidence that spouses' disagreement drives their distributional concerns within households (Cosaert and Lu, 2026). Second, spouses' bargaining power may shift over time when

commitment is limited, which would in turn affect fertility and the division of labor, a channel that the fixed bargaining weights in this paper rule out. Third, the cost of children also depends on investment in their quality, which in East Asia is driven by fierce education competition. Studying this channel would complement this paper by showing how investment in child quality relates to the division of labor between spouses.

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A Appendix

A.1 Data and variables

A.1.1 Gender Value Index (GVI)

Figure A1 reports the distribution of the observed Gender Value Index (GVI) across survey years from 2011 to 2021, showing the sample shifts gradually toward more progressive views over time.[29]

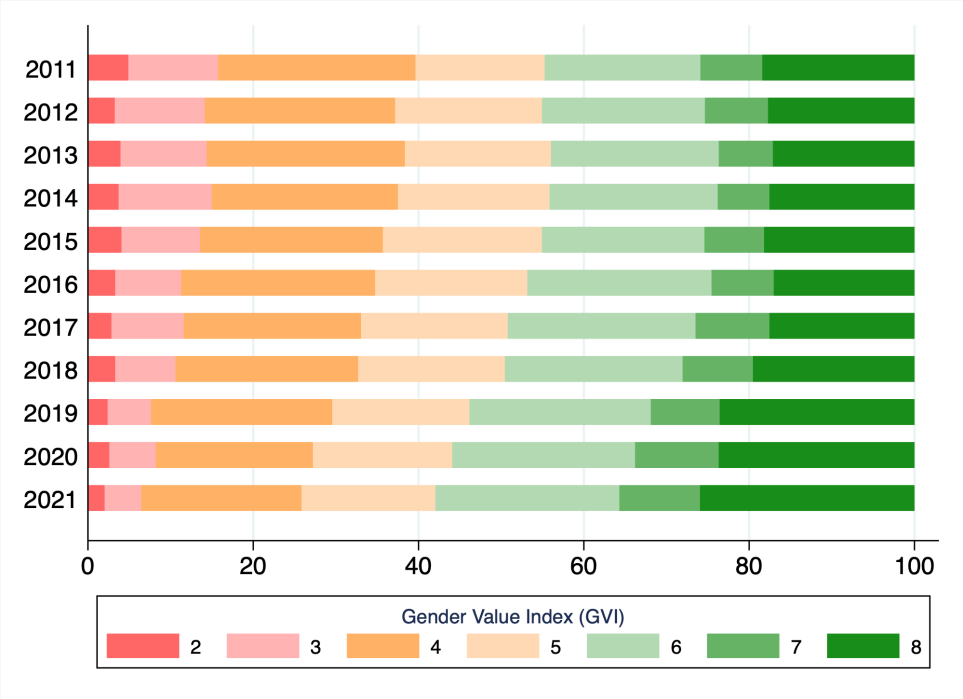


Figure A1: Distribution of Gender Value Index (GVI) by Year

Table A1: Within-Individual Change in GVI with Age

	GVI
Female's age	0.023***
	(0.006)

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

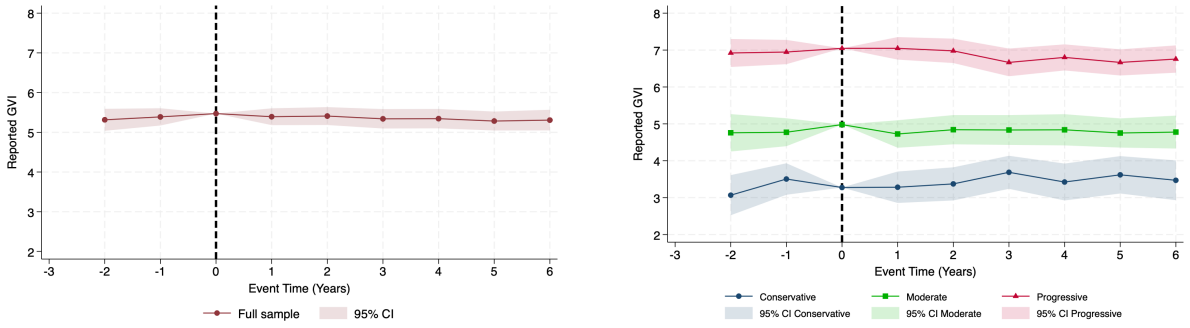
Despite this evolving trend, individual GVI remains highly stable across years. Controlling for individual fixed effects, Table A1 indicates that GVI rises by only 0.023 points per additional year of age, so on the 2–8 scale it would take more than 40 years for the same woman's index to increase by one unit.

[29]The sample shown in this figure includes all female respondents—both single and married—for the years 2011–2021 where the GVI is reported, without applying the sample selection criteria.

I further examine whether observed GVI changes around major life events, controlling for individual and year fixed effects in Table A2. Columns 1–5 report the estimated changes in GVI associated with marriage, divorce, and the births of a first, second, and third child, respectively, and Column 6 includes all these events jointly. None of the coefficients is statistically significant, providing no evidence that individual gender role attitudes respond to these life events. The bottom panel reports the standard deviation of the individual effects (σ_u), the standard deviation of the idiosyncratic error (σ_e), and the intraclass correlation (ρ). These statistics indicate that approximately 66% of the variance in observed GVI is explained by persistent differences across individuals, with the remainder attributable to within-individual fluctuations over time.

To address concerns about anticipation effects and temporary changes around life events, Figure A2 uses the first childbirth as an example and traces the time trend of GVI around it using an event-study design. As discussed in the main text, GVI shows no discernible jump at the time of birth, both in the full sample and within each gender view group. The estimates for each group remain mainly within the range that defines the group. This evidence further supports the stability of self-reported GVI and the consistency of the group classification.

Overall, although individuals' reported gender values fluctuate across waves, their GVI remains largely stable over time. Accordingly, I use each woman's time-averaged GVI to represent her gender role attitudes. This approach enlarges the sample used in the main analysis while reducing the influence of short-run reporting noise.



(a) Full sample

(b) By attitude group

Figure A2: Gender Value Index around the first childbirth

Notes: The sample includes all female respondents, both single and married, for the years 2011–2021 in which the GVI is reported. The figure plots event-time coefficients from a regression of the self-reported Gender Value Index (GVI, 2–8 scale) on event-time dummies relative to the birth of the first child, controlling for individual and year fixed effects. Each coefficient measures the change in GVI relative to the birth year. Shaded areas show 95% confidence intervals based on standard errors clustered at the individual level. In panel (b), the regression is estimated separately for each group. Women are classified as conservative (below 4), moderate (4 to below 6), or progressive (6 or above) based on their average GVI across all survey years.

I further show that GVI is only weakly associated with education and wages, as a regression of GVI on education and wage yields a very low R^2, reported in Table A3.

Table A2: Gender Value Index and Life Events

	(1) Marriage	(2) Divorce	(3) First Childbirth	(4) Second Childbirth	(5) Third Childbirth	(6) All
Marriage	-0.095					-0.093
	(0.061)					(0.061)
Divorce		0.101				0.099
		(0.097)				(0.097)
First childbirth			0.080			0.078
			(0.058)			(0.058)
Second childbirth				0.006		0.007
				(0.059)		(0.059)
Third childbirth					-0.066	-0.063
					(0.090)	(0.090)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Between SD	1.425	1.425	1.425	1.425	1.425	1.425
Within SD	1.025	1.026	1.025	1.026	1.026	1.025
ρ	0.659	0.659	0.659	0.659	0.659	0.659
Observations	16192	16192	16192	16192	16192	16192

Notes: Standard errors in parentheses. *** 1%, ** 5%, * 10%. The sample includes all female respondents—both single and married—for the years 2011-2021 where the GVI is reported, without applying the sample selection criteria described in Section 3. The table presents coefficients from a regression of the self-reported Gender Value Index (GVI, 2–8 scale) on event dummies, controlling for individual and year fixed effects.



Figure A3: Distribution of the Mean-Imputed Gender Value Index (GVI) for Married Females

Figure A4 further presents the share of women with different gender role attitudes by education group.

Table A3: Low R^2 of GVI regressed on Education and Wage

	(1) GVI
Educational attainment	0.116***
	(0.0157)
Hourly wage rate	0.465***
	(0.0193)
Constant	4.843***
	(0.0453)
Observations	16192
R^2	0.041

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The sample includes all female respondents—both single and married—for the years 2011-2021 where the GVI is reported, without applying the sample selection criteria described in Section 3. Educational attainment is scaled from 0 to 7, reflecting a detailed educational degree ranging from junior high school to graduate school. Women who are not wage earners have a wage rate of 0.

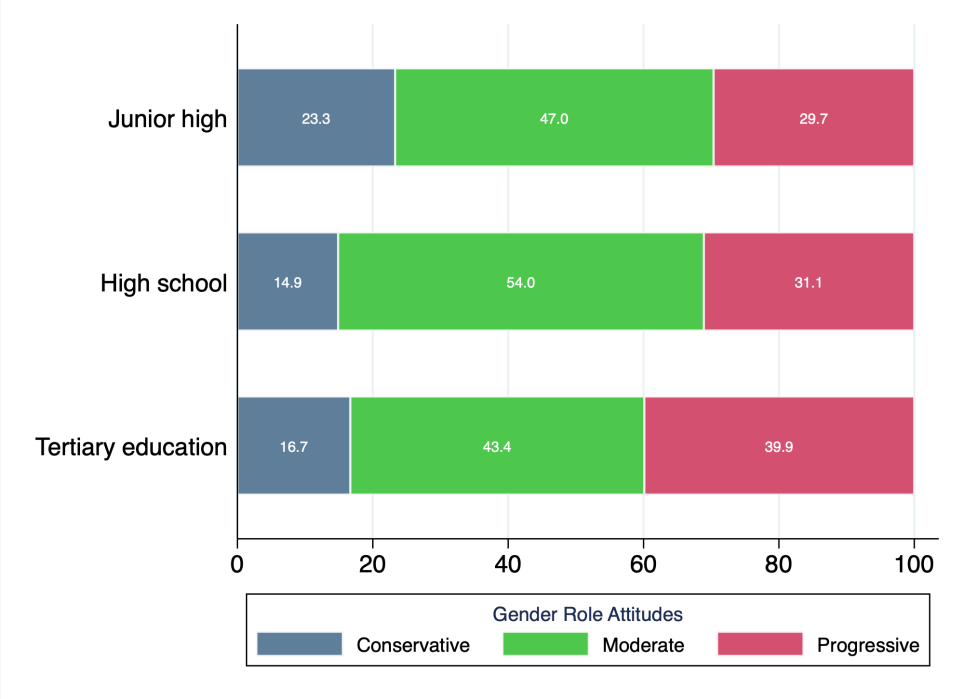


Figure A4: Share of Gender Role Attitudes by Education Group

A.1.2 Stated Birth Intentions

Table A4: Conditions for Additional Childbirth

	Answer: Yes	Answer: No
To be able to pay living expenses with a child or another child	2055 (76.9%)	619 (23.1%)
Cooperation from husband and family	1214 (45.4%)	1460 (54.6%)
To have a larger house	686 (25.7%)	1988 (74.3%)
Available social systems (daycare centers and childcare moms)	640 (23.9%)	2034 (76.1%)
To be able to continue working after giving birth	621 (23.2%)	2053 (76.8%)

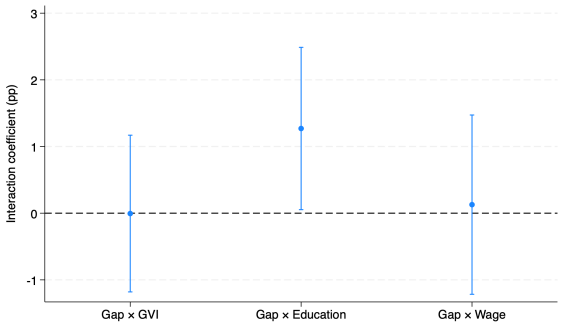
Table A5: Reasons Against Additional Childbirth

	Yes	No
I am satisfied with having my children	5385 (63.8%)	3057 (36.2%)
I am too weak physically to have a child	4559 (49.8%)	4590 (50.2%)
Childcare burden and educational expenses will be heavy	4468 (48.8%)	4681 (51.2%)
Raising a child to become a full member of society is not easy	3329 (36.4%)	5820 (63.6%)
I consider my own life as more important	1498 (16.4%)	7651 (83.6%)
Need to get a larger house	1148 (12.5%)	8001 (87.5%)
I want to continue working	1137 (12.4%)	8012 (87.6%)
I worry whether or not I will have a healthy child	1099 (12.0%)	8050 (88.0%)
I cannot expect that my husband will help me with childcare	941 (10.3%)	8208 (89.7%)
I consider mine and husband's lives as more important	383 (4.2%)	8766 (95.8%)
I don't like children	169 (1.8%)	8980 (98.2%)
I am simply satisfied having my daughter	32 (0.3%)	9117 (99.7%)
I am simply satisfied having my son	15 (0.2%)	9134 (99.8%)

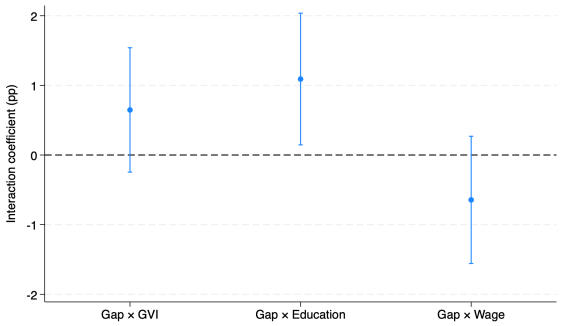
Table A6: Interaction effect of housework gap with women's characteristics

	Condition			Reason		
	(1) Supportive husband	(2) Continue working	(3) Childcare centers	(4) Unsupportive husband	(5) Childcare costs	(6) Continue working
Gap × GVI	0.247	-0.764	-0.334	0.988**	-0.005	0.647
	(0.888)	(0.884)	(0.827)	(0.414)	(0.599)	(0.455)
Gap × Education	-1.076	-0.429	-0.283	-0.206	1.270**	1.091**
	(0.798)	(0.783)	(0.723)	(0.345)	(0.620)	(0.482)
Gap × Wage	-0.235	0.026	-0.923	0.175	0.128	-0.644
	(0.806)	(1.027)	(0.888)	(0.424)	(0.685)	(0.466)
Housework gap	2.904***	-1.102	1.051	1.839***	1.383**	-1.125**
	(0.893)	(0.909)	(0.850)	(0.389)	(0.638)	(0.455)
GVI	0.541	10.270***	3.432	-3.641***	-1.207	-0.284
	(2.825)	(2.976)	(2.687)	(1.318)	(2.295)	(1.834)
Education	5.445*	3.778	-2.858	0.465	-4.058*	-4.969**
	(3.069)	(3.270)	(2.774)	(1.113)	(2.339)	(2.067)
Wage	3.098	1.858	5.140	0.267	-3.367	1.972
	(3.184)	(3.891)	(3.125)	(0.990)	(2.064)	(1.901)
Number of children	-4.003*	-0.003	-1.867	-1.164	4.204***	1.205
	(2.052)	(2.051)	(1.680)	(0.883)	(1.498)	(1.118)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes	Yes	Yes
Male education FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1409	1409	1409	5343	5343	5343

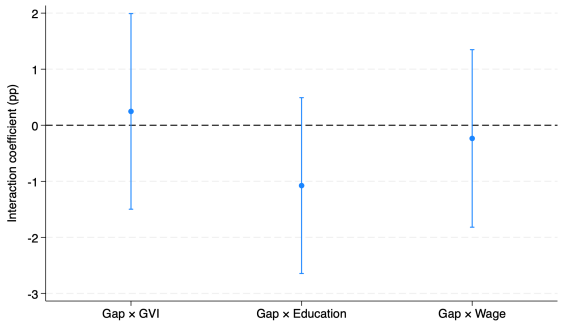
Notes: Linear probability models. Outcomes are in percentage points, and the housework gap is measured in 10 hours per week for easier interpretation. GVI, education, and wage are standardized among all women who answered the intention question. Columns 1–3 use women with a conditional birth intention, and columns 4–6 use women with no birth intention. Standard errors clustered by individual in parentheses. * p<0.10, ** p<0.05, *** p<0.01.



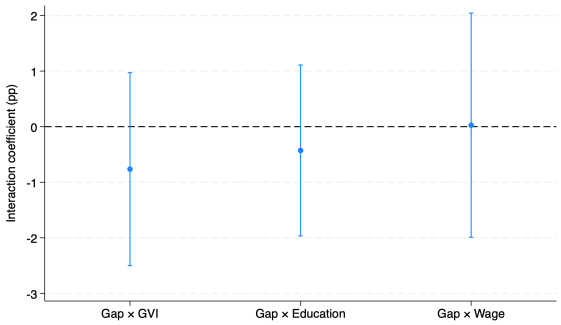
(a) Reason: childcare and education costs



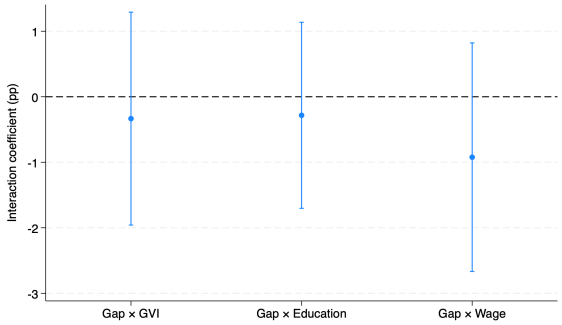
(b) Reason: continue working



(c) Condition: supportive husband



(d) Condition: continue working



(e) Condition: childcare centers

Figure A5: Interaction effects across gender role attitudes, education, and wages

A.2 More empirical evidence

A.2.1 Fertility outcomes and GVI

Table A7: Mean children and number of women by GVI

GVI	Mean births	N	Share (%)
2	1.974	151	2.8
3	2.032	530	10.0
4	1.922	1,249	23.5
5	1.934	982	18.5
6	1.889	1,135	21.4
7	1.861	395	7.4
8	1.791	867	16.3
Total	1.904	5,309	100.0

Note: The reported GVI is the sum of the responses to two questions on gender role attitudes, each scored from 1 to 4, so it takes integer values from 2 to 8.

Table A8: Slope of fertility on reported GVI

	(1) No controls	(2) Age + Education
GVI	-0.036***	-0.027***
	(0.007)	(0.007)
Observations	5,309	5,309
R-squared	0.005	0.045

* p<0.10, ** p<0.05, *** p<0.01.

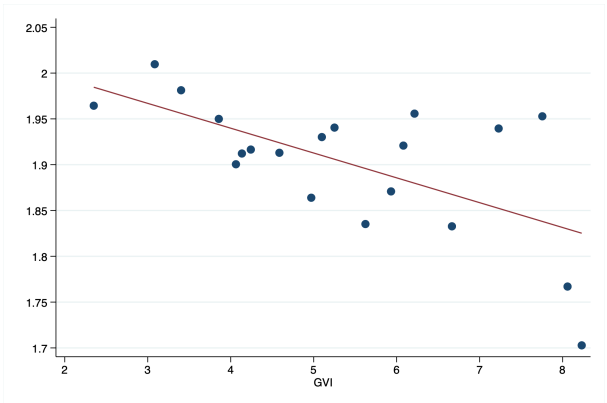


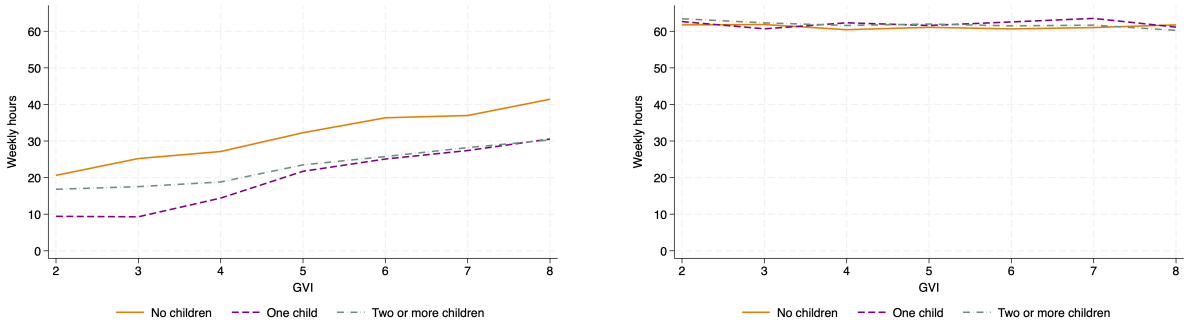
Figure A6: Number of children by GVI, controlled for age and education

A.2.2 Gendered division of labor

Figure A7 presents spouses' time allocation across three groups: childless households, households with one child, and households with two or more children. Spouses' hours are plotted against the wife's GVI on the x-axis for both partners, with the husband's allocation measured against his wife's GVI rather than his own. Panels (a) and (b) show weekly market work hours for wives and husbands, respectively, while Panels (c) and (d) show weekly home production hours.

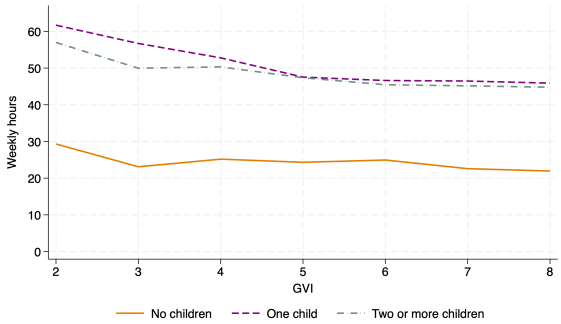
The figure reveals two main patterns. First, women with more progressive gender views devote more time to market work and take on a more balanced share of domestic responsibilities. In panels (a) and (c), wives' market work hours increase with GVI, while their home production hours decline with GVI. Husbands' market work, by contrast, remains largely stable across their wives' gender role attitudes, and their home production hours increase slightly with their wives' GVI, likely reflecting assortative matching on personal values. These patterns are consistent with stronger labor market prospects and greater expectations of within-household equality among more progressive women.

Second, the gendered division of labor is reinforced in households with children. Among childless couples, wives' market work ranges from 20 to 40 hours per week across the GVI distribution, while husbands maintain a stable 60 hours per week. Among households with children, wives' average market work falls by roughly 10 hours per week. Home production hours show a parallel pattern. In childless households, wives spend 20–30 hours per week on home production compared to 0–5 hours for husbands. Once a child is present, wives' home production hours roughly double, rising to 50–60 hours per week, while husbands' home production hours also roughly double, to 5–10 hours per week. The result is a widening spousal gap in housework from the childless to the with-children state, indicating an intensification of traditional gender roles within the household following childbirth.

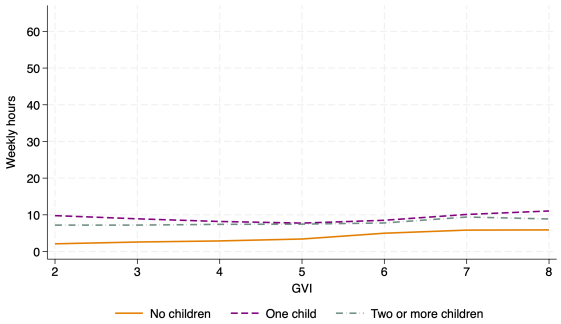


(a) Females' market work hours

(b) Males' market work hours



(c) Females' home production hours



(d) Males' home production hours

Figure A7: Spouses' time allocation by number of children

A.2.3 Event-Study Analysis

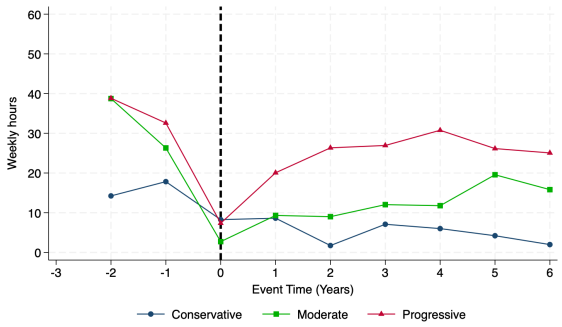
Table A9: Number of women by GVI group and event time

Event time	Conservative	Moderate	Progressive
−2	48	132	132
−1	84	209	181
0	98	249	211
1	98	262	228
2	92	255	208
3	88	238	188
4	84	219	171
5	75	204	165
6	66	193	149
Observations	733	1961	1633

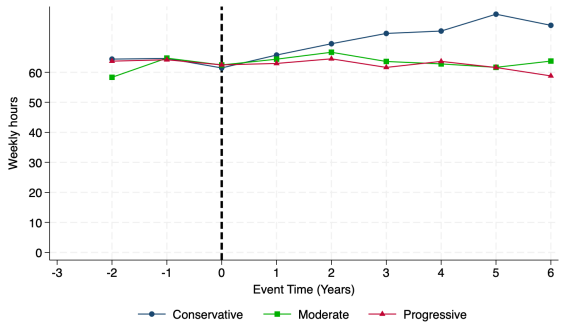
Note: Each cell is the number of distinct women observed at a given event time relative to the first birth. The panel is unbalanced.

Figure A8 presents the effect of childbirth on spouses' time allocation, restricting the sample to couples who had one child and no subsequent births during the observation period. This addresses the concern that more conservative groups tend to have more children in the years following the first birth, which could confound the group comparisons

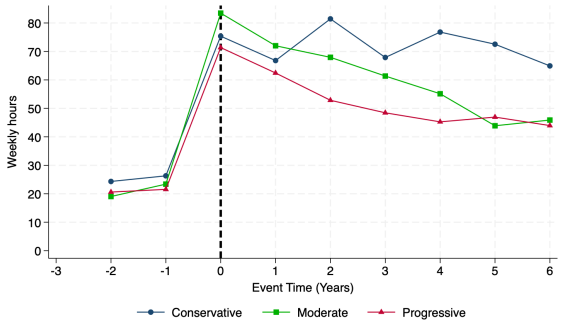
in the full sample. The number of households with the most conservative gender role attitudes and only one child is small—just 39 households—so the event-study estimates for this group are relatively noisy. Nevertheless, the figure shows patterns consistent with the full sample in Figure 3: wives' market work hours and housework hours continue to exhibit pronounced differences across groups.



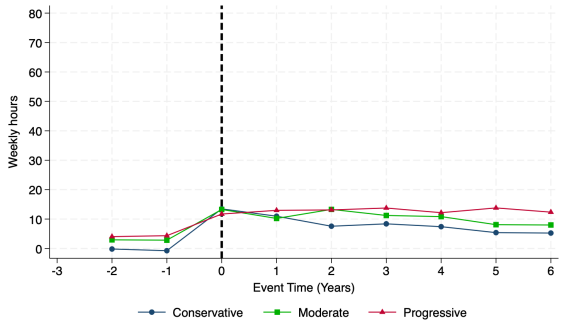
(a) Females' market work hours



(b) Males' market work hours



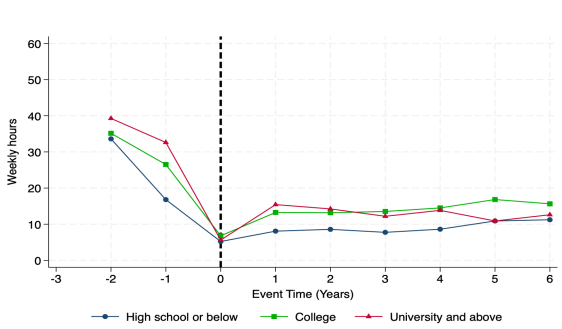
(c) Females' home production hours



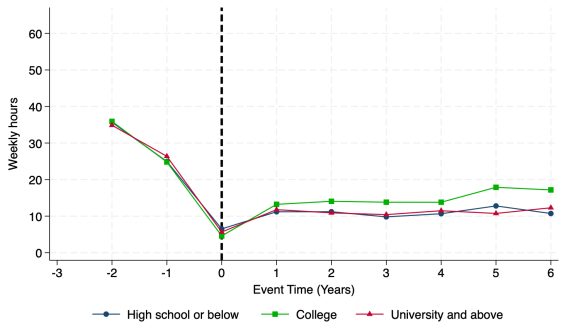
(d) Males' home production hours

Figure A8: Effect of first childbirth on time allocation, among families with one child

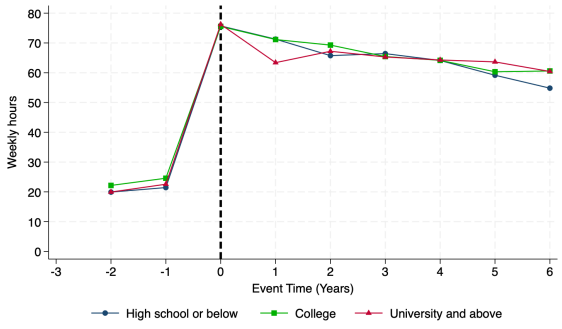
As an additional check that the effects across gender-value groups are not driven by education, I present the effects of childbirth for women with different levels of educational attainment in Figure A9. I consider three groups: agents who completed high school or below, agents with a three-year college degree, and agents with a university or postgraduate degree. I combine junior high and high school into a single group to achieve more even sample sizes across categories. Unlike the specification in Equation 1, the control variables used here exclude the individual's own education, though spousal education is still included. All other regressors remain identical to the baseline specification.



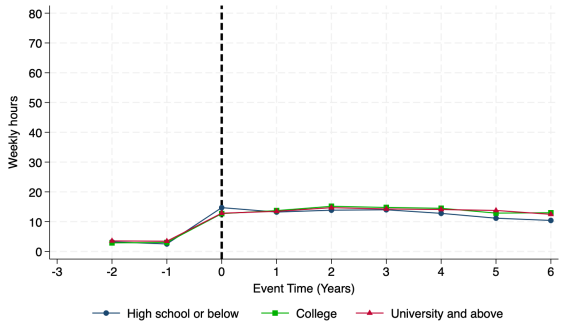
(a) Females' market work hours



(b) Males' market work hours



(c) Females' home production hours



(d) Males' home production hours

Figure A9: Effect of first childbirth on time allocation, by education

A.3 Conception Probability

The probability of conception declines with age, consistent with medical evidence and prior work on fertility. This decline reflects two underlying components, biological fecundity and the risk of permanent infertility. The literature typically calibrates age-dependent fecundity profiles based on medical evidence, but differs in the treatment of infertility risk.

Among the closest literature, Adda et al. (2017) assume women do not anticipate permanent infertility risk and fix 5% of women as permanently infertile ex-ante, with no age variation. Some other papers treat permanent infertility as an absorbing state by allowing the share of infertile women to increase with age, and assume women fully anticipate this risk when making decisions (Sommer, 2016; Jakobsen et al., 2025). [30]

My approach takes a middle ground. I allow the permanent infertility risk to increase with age and occur for individual women stochastically each period, but assume women do not anticipate their individual infertility risk when making decisions. As a result, women are slightly over-optimistic about their individual conception probability. However, the population-level birth probability used in the model solution is correctly calibrated as the product of the biological fecundity rate and the share of fertile women at each age,

[30] Jakobsen et al. (2025) additionally incorporate unplanned births, whereas this paper, in line with the rest of the literature, abstracts from this margin to facilitate identification.

so the aggregate fertility dynamics are consistent with the data.

Let $f_t = 1$ indicate that a woman is permanently infertile and $f_t = 0$ that she is fertile. Let $e_t = 1$ indicate that a couple is trying to conceive and $e_t = 0$ otherwise. The total probability of a birth occurring next period for couples trying to conceive is given by:

$$\begin{aligned} P(b_{t+1} = 1 \mid e_t = 1) &= P(b_{t+1} \mid e_t = 1, f_t = 0)P(f_t = 0) + \underbrace{P(b_{t+1} \mid e_t = 1, f_t = 1)}_{=0} P(f_t = 1) \\ &= P(b_{t+1} \mid e_t = 1, f_t = 0) \cdot P(f_t = 0) \end{aligned} \tag{19}$$

Since permanently infertile women cannot conceive, their contribution to the birth probability is zero, and the total probability reduces to the product of two components: the biological fecundity rate $P(b_{t+1}|e_t = 1, f_t = 0)$ and the share of fertile women at age t: $P(f_t = 0)$.

The fecundity rate measures the probability of conception for a fertile woman trying to conceive within one year. I calibrate this from Leridon (2004), who reports the following age-specific rates: 90% at age 20, 75% at age 30, 66% at age 35, and 44% at age 40. I additionally assume that fecundity is 0% at age 45 and above, and use linear interpolation to construct the complete age profile from these data points.

The share of permanently infertile women increases with age. Drawing on Sommer (2016) and Trussell and Wilson (1985), I use the following age-specific rates of permanent infertility: 1% at age 25, 3% at age 30, 7% at age 35, 17% at age 40, and 97% at age 45 and above. The share of fertile women at age t is then $P(f_t = 0) = 1 - P(f_t = 1)$. As with fecundity, I use linear interpolation to obtain the complete age profile.

I assume infertility is an absorbing state, meaning a woman remains permanently infertile once she becomes infertile. The share of infertile women evolves as:

$$\begin{aligned} P(f_{t+1} = 1) &= P(f_{t+1} = 1 \mid f_t = 0)P(f_t = 0) + \underbrace{P(f_{t+1} = 1 \mid f_t = 1)}_{=1} P(f_t = 1) \\ &= P(f_{t+1} = 1 \mid f_t = 0)(1 - P(f_t = 1)) + P(f_t = 1) \end{aligned} \tag{20}$$

This transition probability enters the model each period as an exogenous risk faced by fertile women.

A.4 Grid Points

Table A10: Preset grid points for the model variables

Variable	Value
Bargaining power	[0.48, 0.72, 0.98, 1.38]
Number of children	[0, 1, 2, 3]
Age of youngest child	[0, 2, 4, 6, 8, 10]
Females' weekly work hours	[0, 25, 40, 55]
Males' weekly work hours	[60]
Human capital stock	[0,1,2,3]
Assets	[0, 48.9, 196.6, 446.4, 637.7]
Equivalence scales of child-related expenses: conservative	[1, 1.52, 2.03, 2.24]
Equivalence scales of child-related expenses: moderate	[1, 1.53, 2.09, 2.48]
Equivalence scales of child-related expenses: progressive	[1, 1.54, 2.15, 2.32]
Female sleep time by number of children: conservative	[85, 76, 77, 75]
Female sleep time by number of children: moderate	[83, 76, 76, 76]
Female sleep time by number of children: progressive	[79, 73, 75, 71]
Male sleep time by number of children: conservative	[78, 73, 75, 74]
Male sleep time by number of children: moderate	[77, 75, 75, 76]
Male sleep time by number of children: progressive	[75, 73, 75, 72]

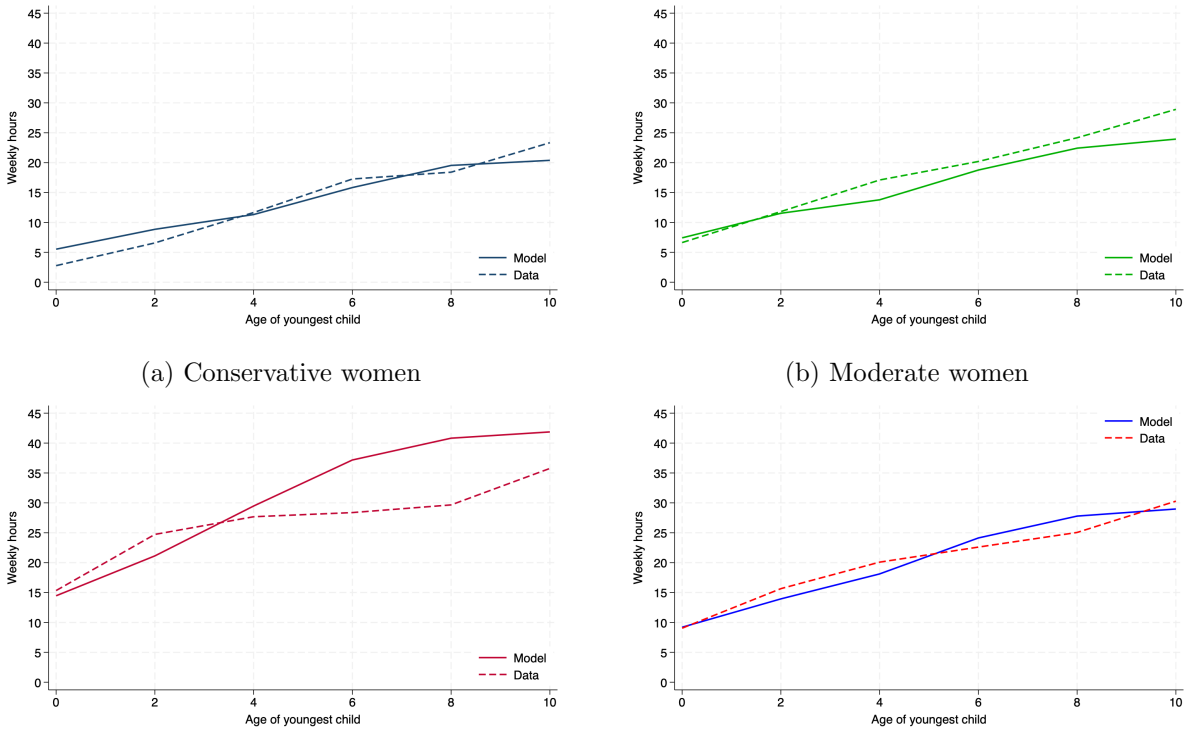
A.5 Initial Distribution

The distribution of the number of children at the initial period varies by female gender role attitudes. Among conservative women, 31.58% have no children at age 24-25, 47.37% have one child, 18.42% have two children, and 2.63% have three or more children. Among moderate women, the corresponding shares are 33.65%, 35.58%, 27.88%, and 2.88%. Among progressive women, the shares are 37.88%, 39.39%, 21.21%, and 1.52%. I randomly draw each household's number of children at period 1 from the distribution corresponding to the wife's gender role attitudes.

The relative wage ratio, which defines the bargaining power states in the model, is discretized into four grid points corresponding to the sample quartiles. Specifically, 25% of women have a relative wage of 0.48 or below, 50% fall below 0.72, 75% fall below 0.98, and the top quartile reaches 1.38. These cutoffs are taken directly from the data. Each agent's wage rate, in turn, depends on their human capital stock, described next.

Agents' initial human capital stock is imputed from a wage regression estimated separately for each gender on the sample of wage earners. Specifically, I regress the log wage rate on education, work experience, working industry, and region of residence. I then use the estimated coefficients to compute each individual's predicted wage, including those not currently employed, based on these same characteristics. For non-employed women, the predicted wage represents their potential wage if they were to work. The predicted wages are then discretized into four quartile bins, which serve as a proxy for human capital stock. Work experience represents potential working years, computed as an individual's age minus years of schooling and six pre-schooling years. Agents' initial human capital follows the empirical distribution of these quartile bins by gender. Among females, 57.21% fall into the lowest bin, 23.08% into the second, 13.94% into the third, and 5.77% into the highest. For males, the corresponding shares are 64.90%, 22.12%, 8.17%, and 4.81%.

A.6 Supplementary model fits



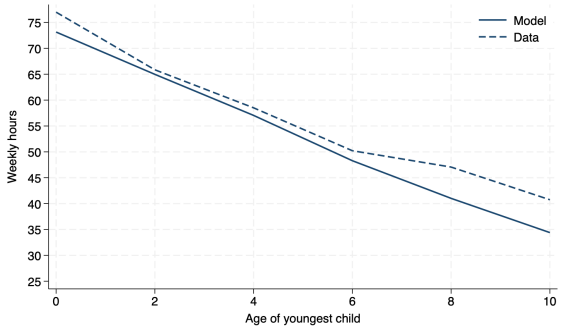
(a) Conservative women

(b) Moderate women

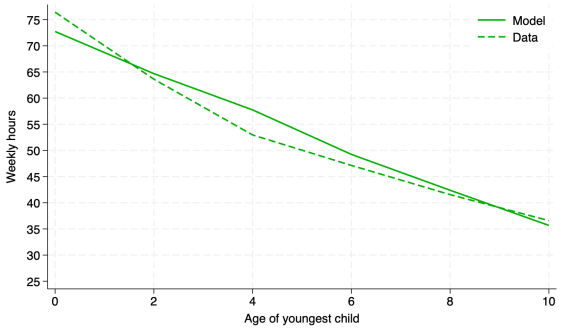
(c) Progressive women

(d) All women

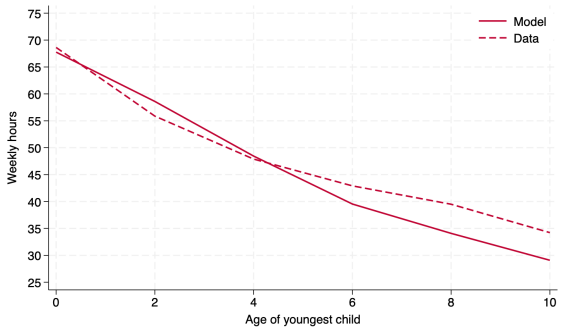
Figure A10: Female market work hours across age of youngest child



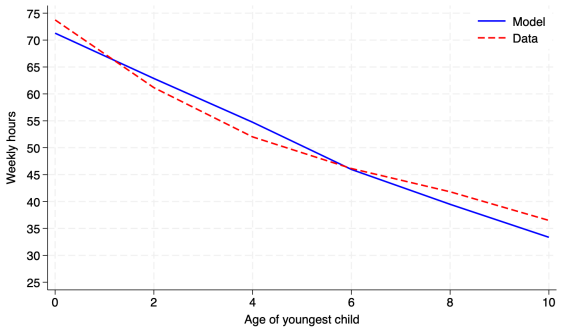
(a) Conservative women



(b) Moderate women

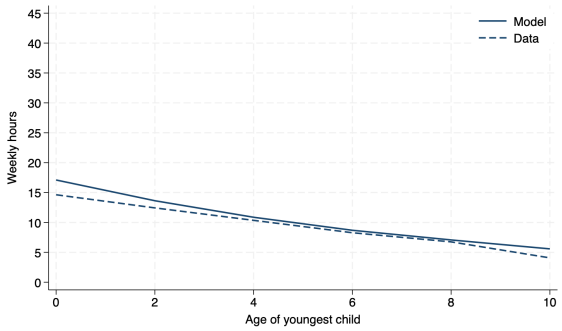


(c) Progressive women

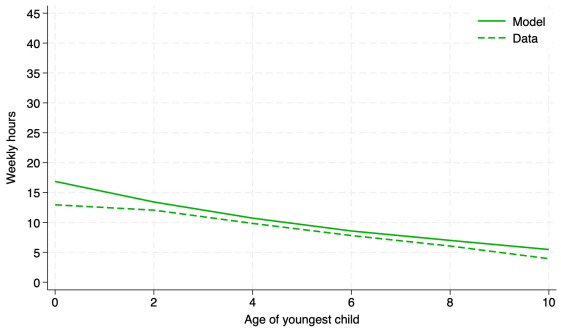


(d) All women

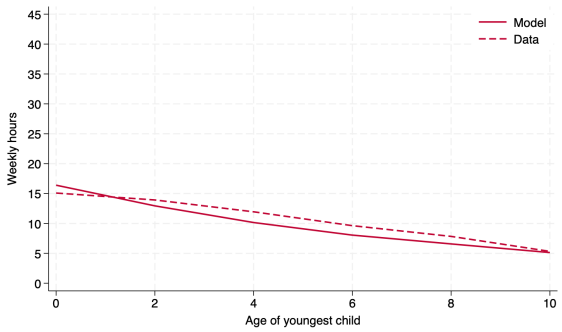
Figure A11: Female home production hours by age of the youngest child



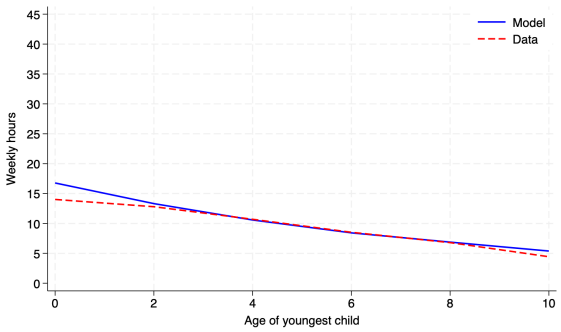
(a) Husbands of conservative women



(b) Husbands of moderate women

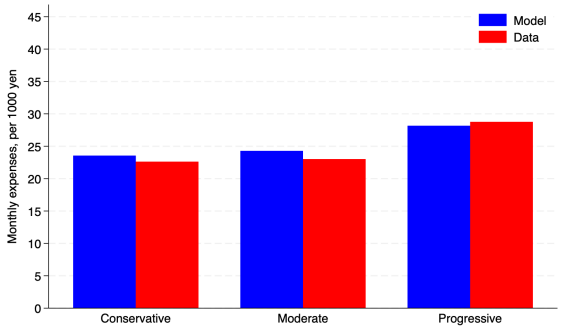


(c) Husbands of progressive women

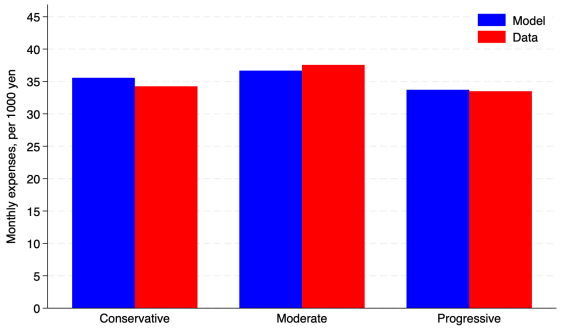


(d) All men

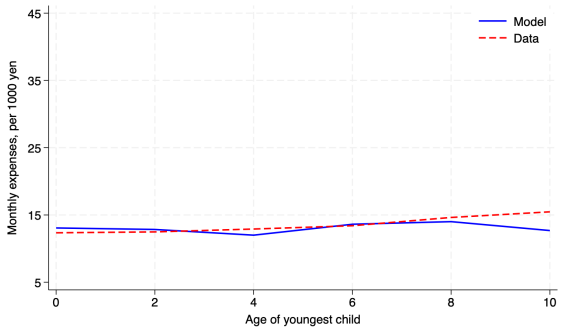
Figure A12: Male home production hours by age of the youngest child



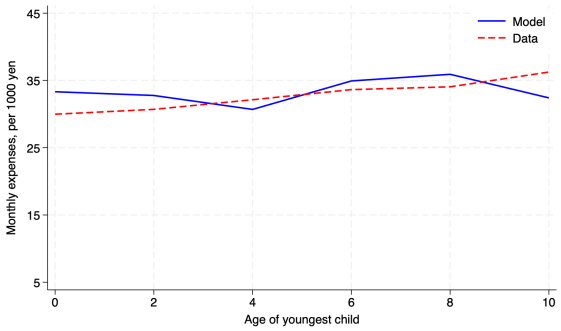
(a) Women's expenses in childless households



(b) Men's expenses in childless households



(c) Women's expenses by age of the youngest child



(d) Men's expenses by age of the youngest child

Figure A13: Untargeted moments: consumption

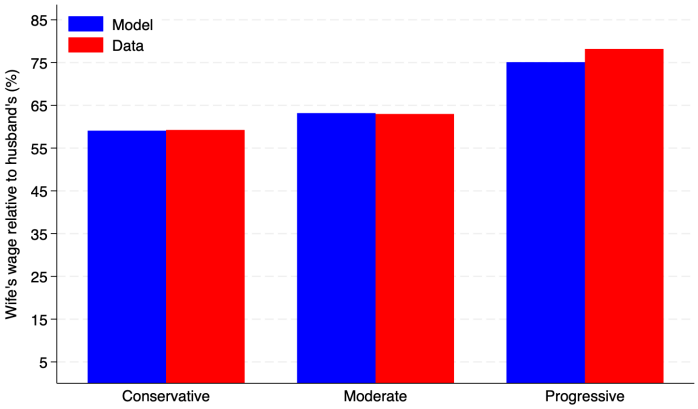


Figure A14: Spouses' relative wage across gender view groups

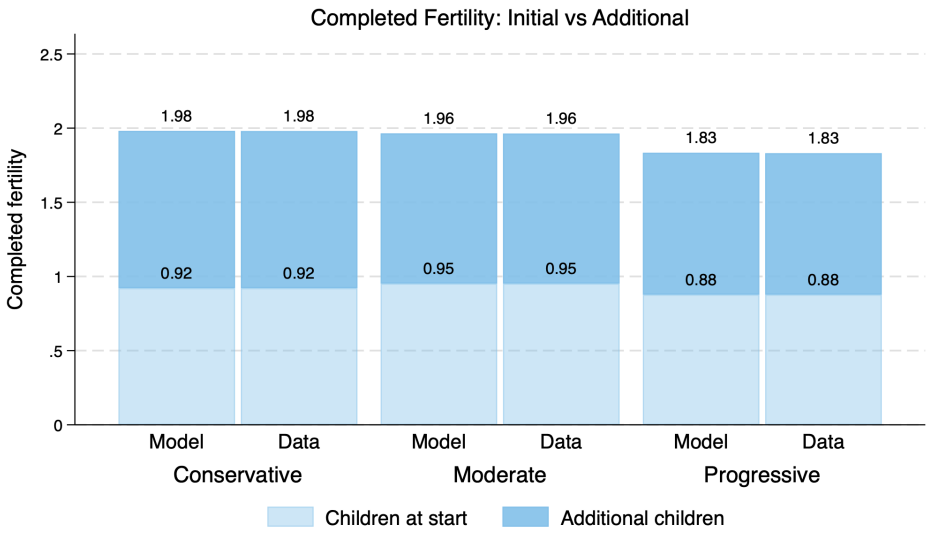
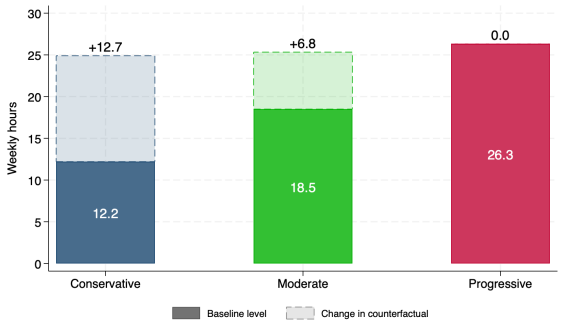
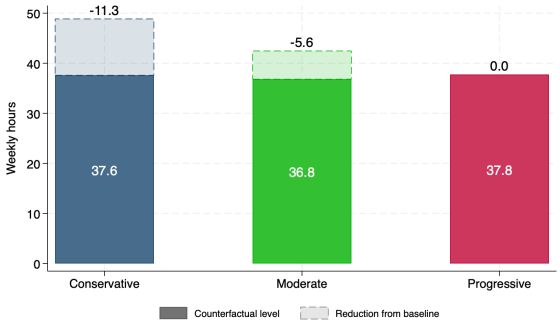


Figure A15: Initial versus completed number of children

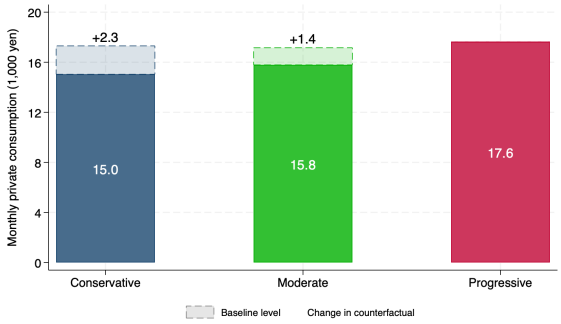
A.7 Supplementary counterfactual results



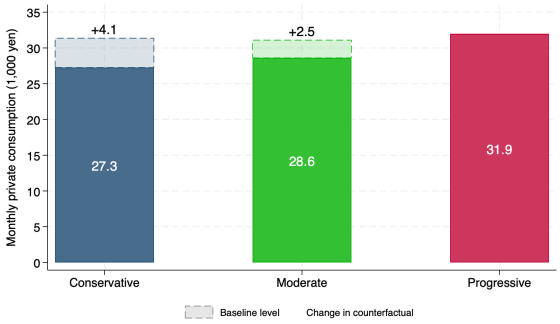
(a) Women's market work



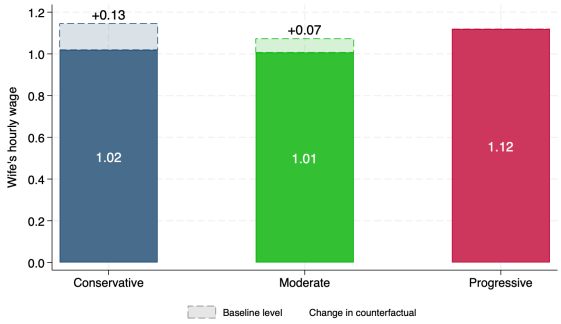
(b) Women's home production



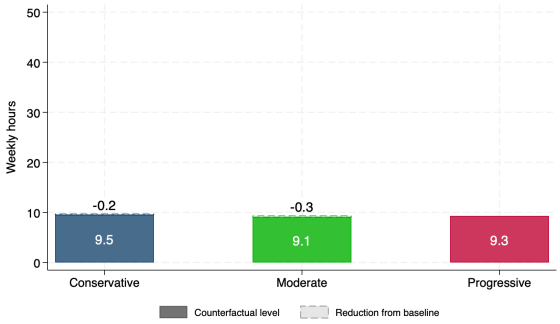
(c) Women's private consumption



(d) Men's private consumption

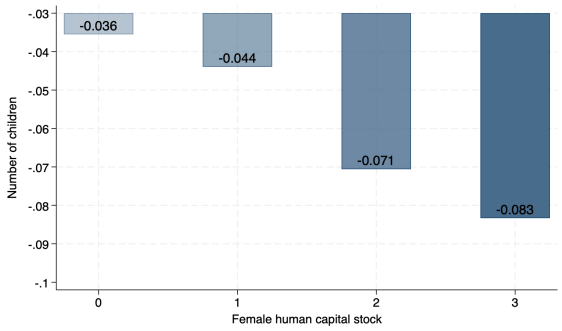


(e) Women's hourly wage

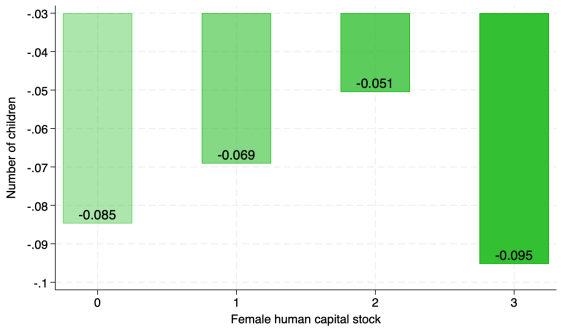


(f) Men's home production

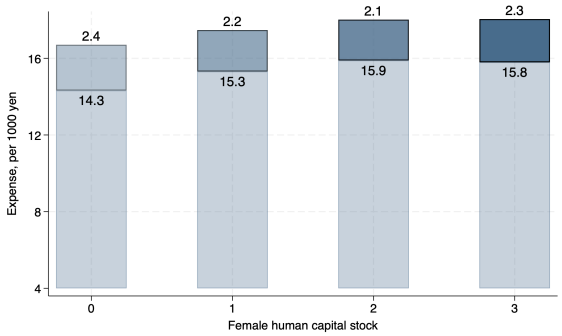
Figure A16: Intra-household outcomes if all women hold progressive views



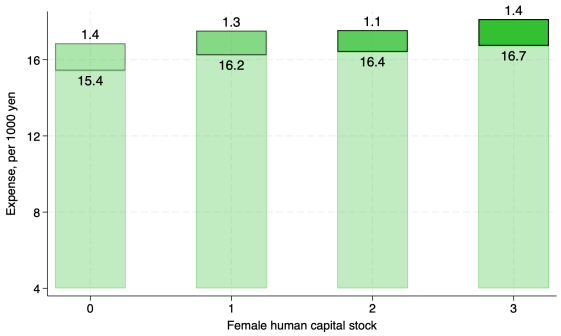
(a) Conservative women: completed fertility



(b) Moderate women: completed fertility



(c) Conservative women: consumption



(d) Moderate women: consumption

Note: The framed segments in plots c–d show the simulated increment from the counterfactual, relative to the baseline values shown in the unframed segment below.

Figure A17: Changes by human capital stock, level

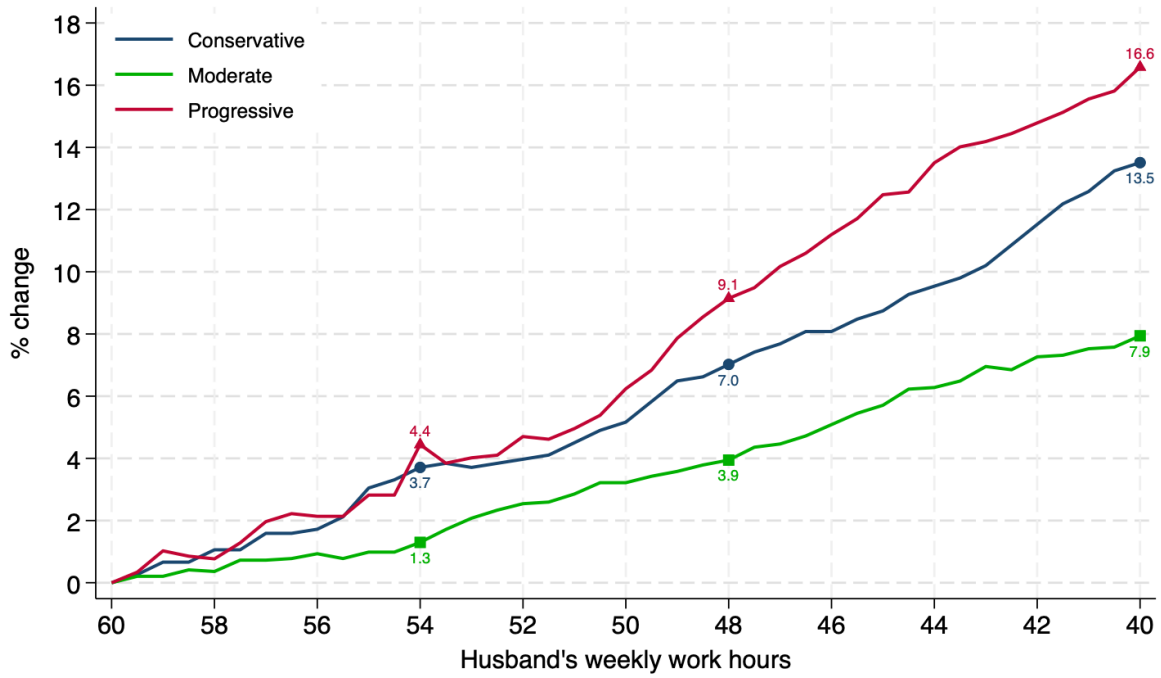
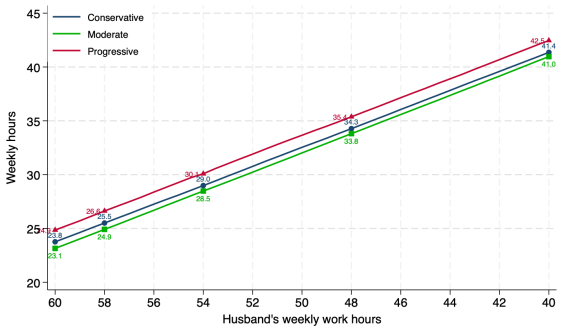


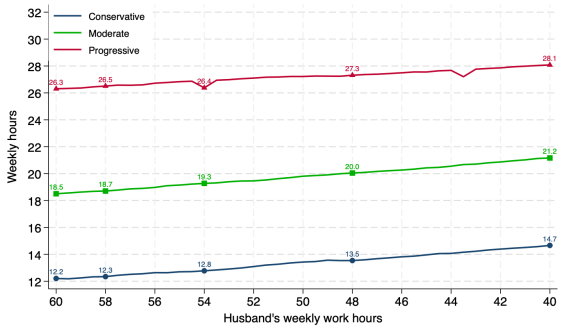
Figure A18: Completed fertility change (%) if husbands reduce market work



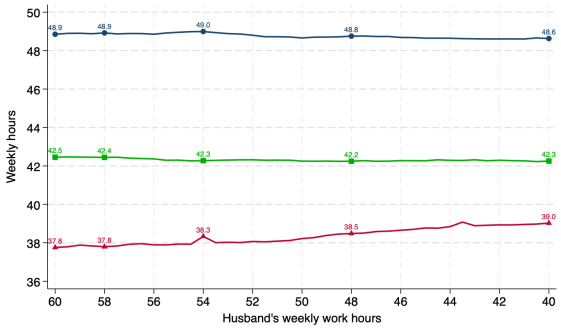
(a) Men's home production



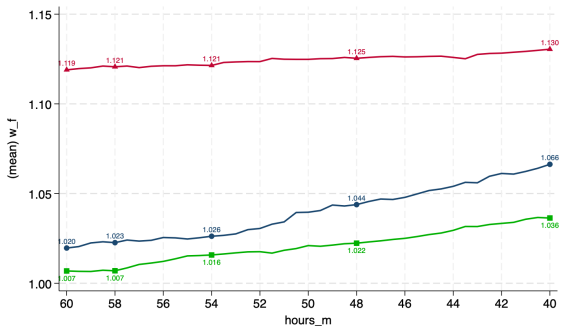
(b) Men's leisure



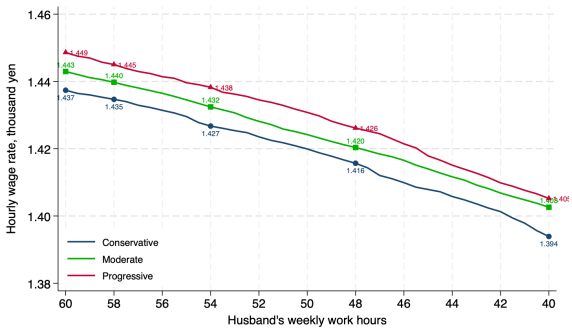
(c) Women's market work



(d) Women's home production

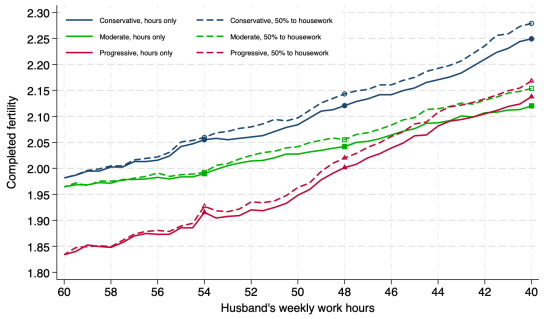


(e) Women's hourly wage

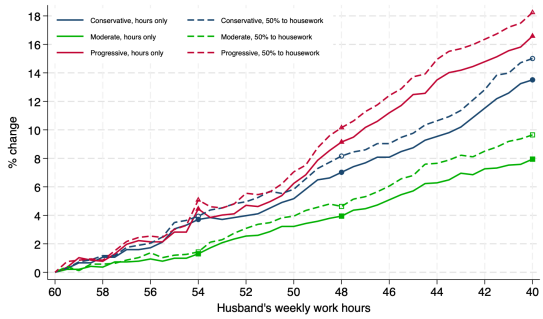


(f) Men's hourly wage

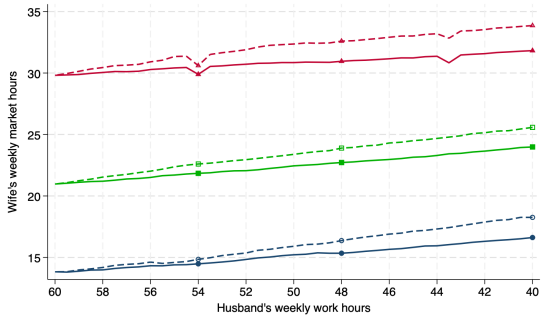
Figure A19: Responses to reduced market hours for husbands



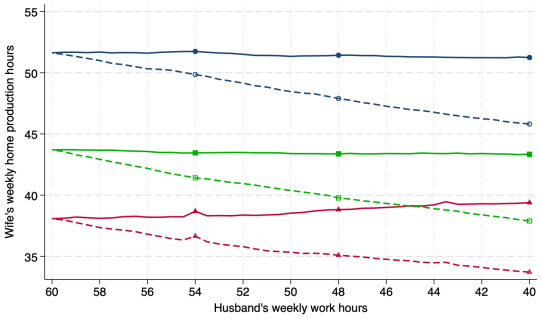
(a) Completed fertility



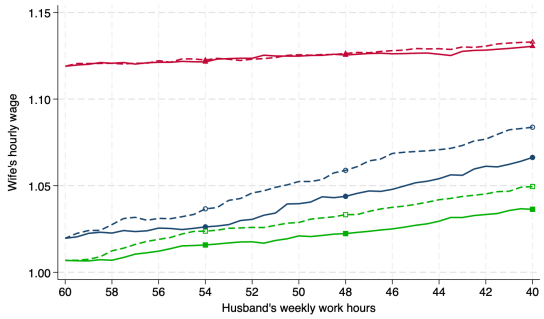
(b) % change in completed fertility



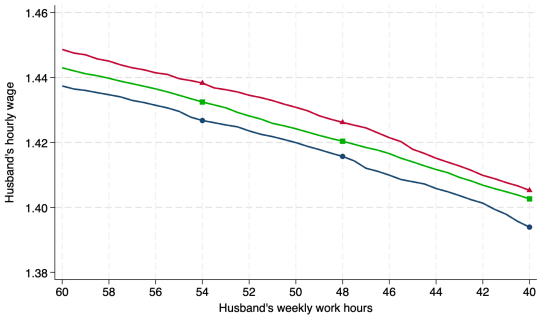
(c) Women's market work



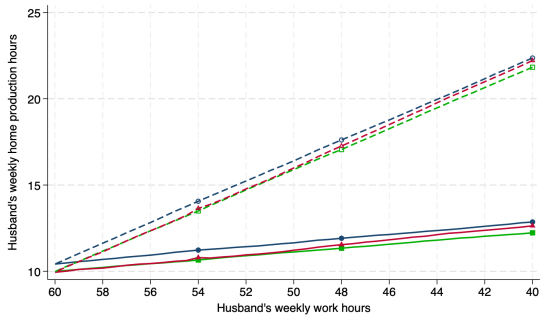
(d) Women's home production



(e) Women's hourly wage, thousand yen



(f) Men's hourly wage, thousand yen



(g) Men's home production



(h) Men's private leisure

Figure A20: Husbands reduced market hours with 50% reallocated to home production